Stellar Atmospheres

Course B of two · Five concept-based, illustrated lectures: the state of the gas, opacity, rates and non-LTE, the gray atmosphere and lines, model atmospheres

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Final Examination — Course B: Stellar Atmospheres (with complete solutions)
Part A — Short answers (2 marks each)
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Formula Sheet — Course B: Stellar Atmospheres
LTE (Lecture SA1)
Continuous opacity (Lecture SA2)
Transition rates (Lecture SA3)
Line absorption and broadening (Lecture SA2)
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Stellar atmospheres (Lecture SA5)

Lectures SA1–SA5, final exam (SA6) and formula sheet (SA7). Prerequisite: Course A, *Radiative Transfer* (RT1–RT6). Every lecture opens with a concept map and ends with misconceptions, a summary and four exam questions with answers.

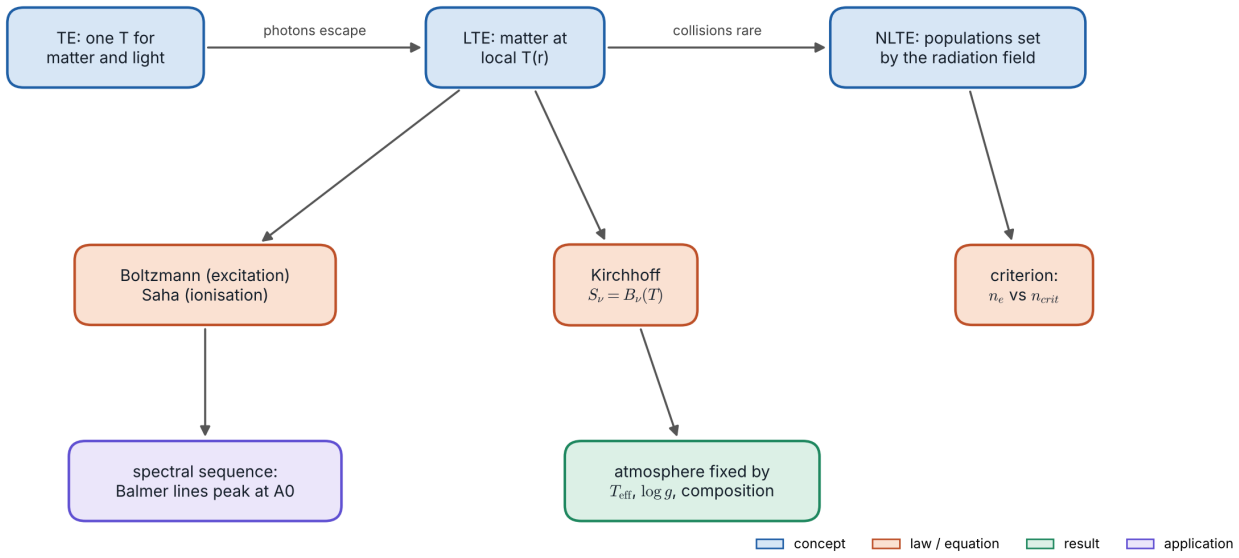
Based on: E. Novotny, *Introduction to Stellar Atmospheres and Interiors* (1973) · D. Mihalas, *Stellar Atmospheres* (1978) · G. W. Collins II, *Fundamentals of Stellar Astrophysics*, Part II · D. F. Gray, *The Observation and Analysis of Stellar Photospheres* · S. Chandrasekhar, *Radiative Transfer* (1950) · Hubeny & Mihalas, *Theory of Stellar Atmospheres* (2014)

Code: figures and numbers come from `scripts/lte_solvers.py`, `scripts/sa_physics.py` and `scripts/worked_examples.py`.

Lecture SA1 — The State of the Gas: TE, LTE and Non-LTE; Saha and Boltzmann

> **Sources:** Collins §9.1 (breakdown of strict TE, assumption of LTE, the five assumptions of normal stellar atmospheres), §9.2c–d (Kirchhoff's law, $S_\nu = B_\nu$); E. Novotny, *Introduction to Stellar Atmospheres and Interiors* (1973), Ch. 3.1–3.3 (Bohr atom, excitation, ionisation); MIT 8.901 §7.3; Mihalas, *Stellar Atmospheres* (1978) Ch. 5; R. N. Thomas (1965); Osterbrock & Ferland (2006).

SA1 concept map — the state of the gas



*Concept map — three regimes, ordered by how much the radiation is allowed to escape. In **TE** one temperature rules everything. In **LTE** photons escape but collisions still set the atoms by the local temperature; Boltzmann, Saha and Kirchhoff then give the opacity and $S_\nu = B_\nu$. In **NLTE** collisions are too rare, and the radiation field takes over.*

Learning goals

1. Say what a stellar atmosphere is, and which assumptions define a "normal" one.
2. Define TE, LTE and NLTE precisely, with two examples of each, and decide which applies.
3. Derive Kirchhoff's law and $S_\nu = B_\nu(T)$ in LTE.
4. Use the Boltzmann and Saha equations, and explain why Balmer lines peak in A stars.

The big ideas in plain words

The atmosphere is where photons escape. It is the thin skin of the star, less than 0.1 % of the Sun's radius, from where the gas is still opaque ($\tau \sim 10$–100) out to where it is transparent. Yet it produces the entire spectrum we observe.

Three levels of equilibrium — a crowded hall (Figure 1.1). In **TE** the hall is closed and has been quiet for a long time: people (matter) and the air (radiation) share one temperature everywhere. In **LTE** the doors are open and air

streams out (photons escape), but people bump into each other so often that each corner keeps its own local temperature. In **NLTE** the hall is so empty that people rarely meet; what they do is decided by what blows in through the doors, i.e. the radiation field.

Who talks to the atoms decides the regime. Atoms are excited and de-excited by *collisions* (which "know" the local temperature) and by *photons* (which may come from far away, or escape). If collisions win, populations follow the local temperature: LTE. If photons win, populations follow the light: NLTE.

Boltzmann is a tax on climbing the energy ladder. Occupying a level of energy ΔE costs a factor $e^{-\Delta E/kT}$, a heavy tax when kT is small.

Saha balances cost against freedom. Ionisation costs a large energy χ, but a freed electron gains an enormous number of possible states. That freedom wins early: hydrogen ionises near 10 000 K although $\chi/k = 158\,000$ K. Low density helps too, since a free electron rarely meets an ion.

1.1 What a stellar atmosphere is

Collins (§9.1) defines the atmosphere by *the failure of the assumptions that work in the interior*: near the surface photons start to escape, the radiation stops being Planckian, and the transfer of radiation becomes global instead of local. The questions this course answers:

Question	Lecture
What is the state of the gas?	SA1 (this lecture)
How does matter absorb and emit, in the continuum and in lines?	SA2
What sets the level populations when LTE fails?	SA3
What structure and spectrum result in the simplest (gray, LTE) case?	SA4
How are real model atmospheres built, and how does the Sun compare?	SA5

Collins' five assumptions of a "normal" atmosphere. (1) LTE. (2) The structure is set by the continuum opacity. (3) Plane-parallel geometry. (4) Constant gravity with hydrostatic equilibrium, $dP/dz = -\rho g$. (5) Radiative equilibrium, $F = \sigma T_{\rm eff}^4$ = constant. Then **only** $T_{\rm eff}$**,** $\log g$ **and the composition** specify the atmosphere. Lecture SA4 works inside these assumptions; Lecture SA5 shows where each fails. Novotny (1973, Ch. 9) builds a complete solar model by hand under exactly these assumptions; we redo it in Lecture SA5.

1.2 TE, LTE and NLTE: definitions with examples

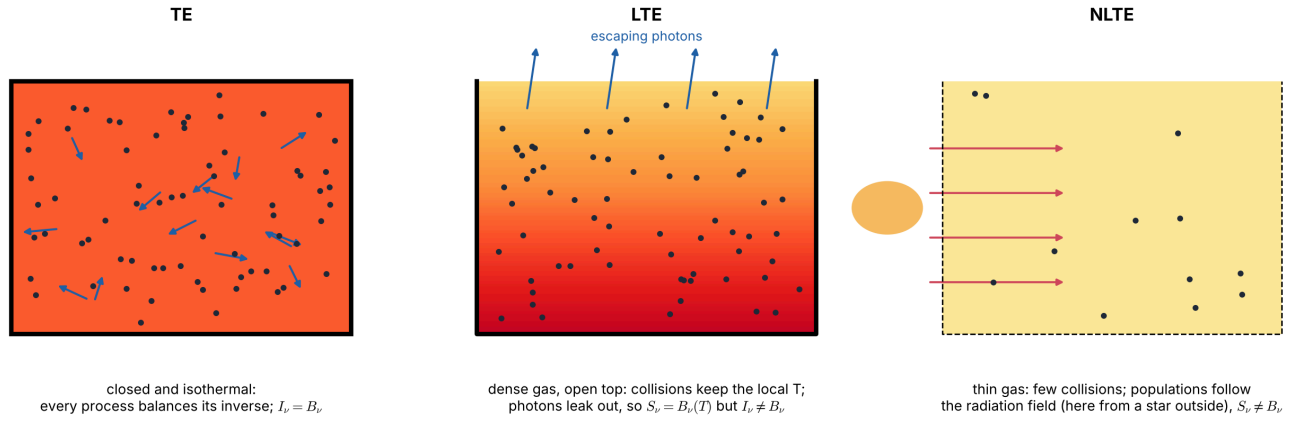


*Figure 1.1 — **TE:** a closed, isothermal box in which every process balances its inverse; $I_\nu = B_\nu$. **LTE:** a dense gas with an open top; collisions keep the local T, photons leak out, so $S_\nu = B_\nu(T)$ but $I_\nu \neq B_\nu$. **NLTE:** a thin gas with few collisions; populations follow the radiation field (here from a star outside), so $S_\nu \neq B_\nu$.*

TE. Every process is balanced by its inverse, and all distributions take their equilibrium form at **one** T: Maxwell, Boltzmann, Saha and Planck, with $I_\nu = B_\nu(T)$ isotropic. Hence $J_\nu = B_\nu$, $F_\nu = 0$, $S_\nu = B_\nu$. *Examples:* a closed oven (the black body); the **solar core**, isotropic to 1 part in 10^{11} (Worked example 1.3); the **early Universe** before recombination, which is why the CMB is Planckian.

LTE. The **matter** at each point is in equilibrium at the **local** $T(\mathbf{r})$ (Maxwell, Boltzmann and Saha at $T(\mathbf{r})$); the **radiation** may differ from B_ν. Then $j_\nu = \alpha_\nu B_\nu$ and $S_\nu = B_\nu(T)$, but $I_\nu \neq B_\nu$ and $J_\nu \neq B_\nu$. *Examples:* the continuum of the solar photosphere and weak metal lines in F, G, K dwarfs (the basis of classical abundance work); Earth's lower atmosphere in the thermal infrared (CO_2 at 15 μm below ~70 km); dense laboratory arcs.

NLTE. Some level populations differ from their Saha–Boltzmann values and must be found from the **rate equations**, which couple them to the radiation field. Then $S_\nu \neq B_\nu(T)$, and the problem is non-local and non-linear. *Examples:* cores of strong solar lines (Na D, Ca II H & K, Hα); chromospheres and coronae; hot O and B stars and their winds; H II regions and planetary nebulae (forbidden lines); molecular clouds; masers (inverted populations, $T_{\rm ex} < 0$); aurorae and airglow.

	TE	LTE	NLTE
temperature	one T	local $T(\mathbf{r})$	kinetic $T(\mathbf{r})$, but it does not set the populations
populations	Boltzmann, Saha (T)	Boltzmann, Saha ($T(\mathbf{r})$)	rate equations: $n_i = b_i\, n_i^{\mathrm{LTE}}$
source function	$S_\nu = B_\nu$	$S_\nu = B_\nu(T(\mathbf{r}))$	depends on J_ν
radiation field	$I_\nu = B_\nu$, $F = 0$	$I_\nu \neq B_\nu$, carries flux	$I_\nu \neq B_\nu$
local or global?	trivial	local: $T \to$ populations $\to$ κ_ν, j_ν	global: populations $\leftrightarrow$ radiation everywhere
criterion	closed, isothermal over many ℓ_γ	collisions $\gg$ radiative rates	radiative rates $\gtrsim$ collisions

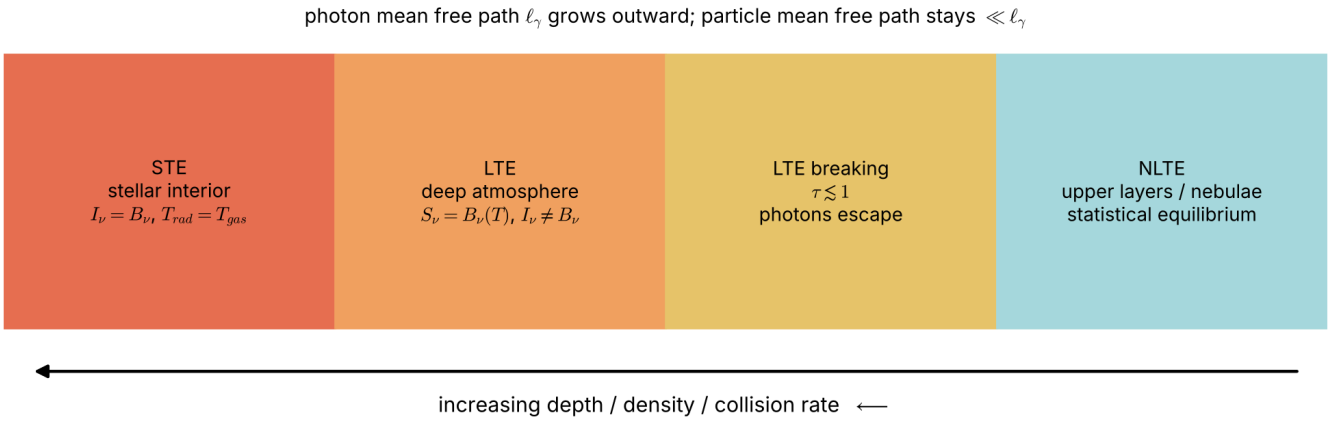


Figure 1.2 — Moving outward through a star: TE in the interior, LTE in the deep atmosphere, NLTE in the upper layers. LTE fails first for strong lines and at low density.

Why the radiation leaves equilibrium first. Photons have much longer mean free paths than particles (Collins §9.1a). Near the surface photons escape and no photons come back from space, so the radiation becomes anisotropic and non-local, while particles still collide often enough to stay Maxwell–Boltzmann–Saha at the local T.

A practical criterion (developed in Lecture SA3). Collisions win for a transition $u \to l$ if $n_e q_{ul} \gg A_{ul}\,\beta_{\mathrm{esc}}$, where $\beta_{\mathrm{esc}} \approx 1/\tau$ is the photon escape probability. Equivalently, compare n_e with the **critical density** $n_{\mathrm{crit}} = A_{ul}\beta_{\mathrm{esc}}/q_{ul}$.

1.3 Kirchhoff's law: why $S_\nu = B_\nu$ in LTE

In a TE cavity $I_\nu = B_\nu$ is uniform and isotropic, so $dI_\nu/ds = 0 = j_\nu - \alpha_\nu B_\nu$:

$$\boxed{j_\nu = \alpha_\nu B_\nu(T)} \quad \Rightarrow \quad \boxed{S_\nu^{\mathrm{LTE}} = B_\nu(T)}$$

Good absorbers are good emitters. j_ν and α_ν depend only on the *state of the matter*. In LTE that state is the TE state at the local T, so the law holds even though $I_\nu \neq B_\nu$. With Boltzmann populations the Einstein formulas of Lecture RT2 give the same result, together with the **LTE stimulated-emission factor** $\left(1 - e^{-h\nu/kT}\right)$ that multiplies every LTE opacity.

True in LTE	*Not* implied by LTE
$S_\nu = B_\nu(T_{\rm local})$	$I_\nu = B_\nu$ or $J_\nu = B_\nu$
populations from Saha–Boltzmann at $T_{\rm local}$	isotropic radiation, zero flux
κ_ν, j_ν are functions of (T, ρ) only	a black-body spectrum (thin gas emits $\tau_\nu B_\nu$; MIT §7.3)

1.4 Boltzmann and Saha

Boltzmann (excitation within one ion):

$$\frac{N_j}{N_i} = \frac{g_j}{g_i} e^{-(E_j - E_i)/kT}, \qquad \frac{N_j}{N} = \frac{g_j e^{-E_j/kT}}{U(T)}, \quad U = \sum_i g_i e^{-E_i/kT}.$$

For hydrogen (Bohr atom, Novotny §3.1), $g_n = 2n^2$ and $E_n = -13.6\ \text{eV}/n^2$.

Saha (ionisation $r \to r+1$):

$$\boxed{\frac{N_{r+1}\, n_e}{N_r} = \frac{2U_{r+1}}{U_r}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_r/kT}}$$

with $(2\pi m_e k/h^2)^{3/2} = 2.415 \times 10^{21}\ \text{m}^{-3}\,\text{K}^{-3/2}$ and $n_e = P_e/kT$. The factor 2 is the electron spin; the $T^{3/2}$ counts free-electron states and makes ionisation happen at $kT \ll \chi$.

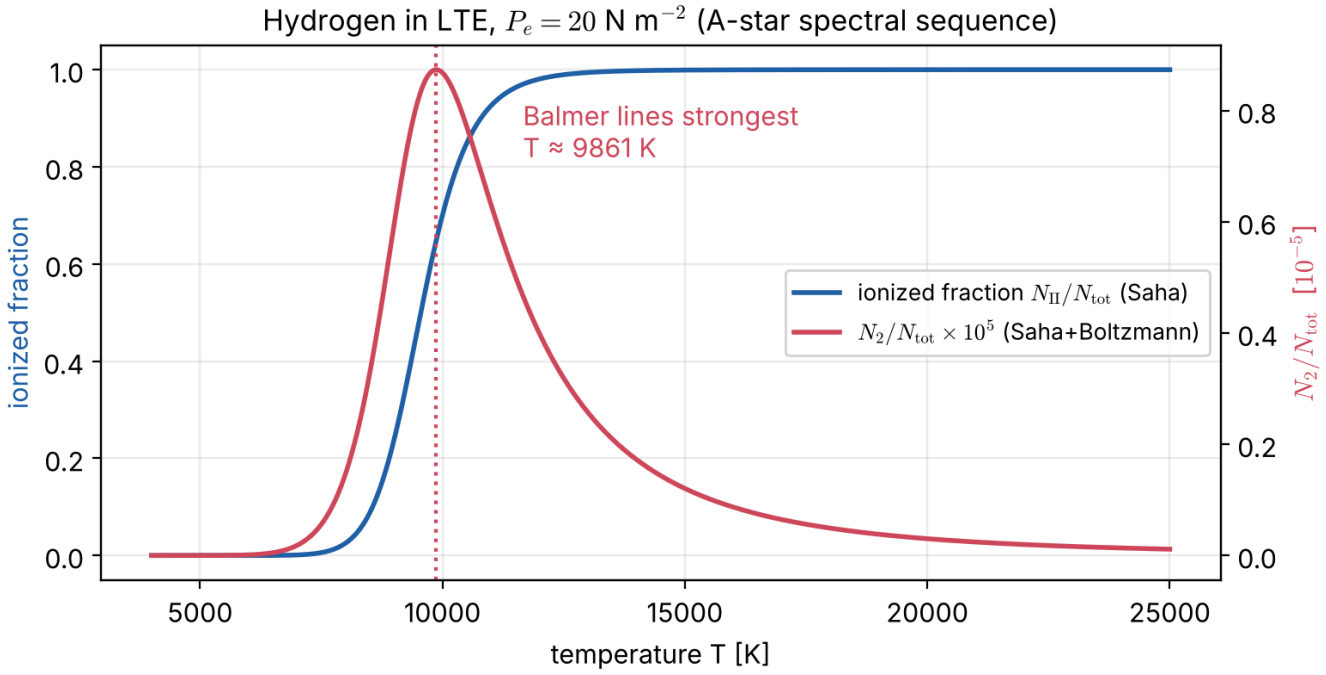


Figure 1.3 — Hydrogen in LTE at $P_e = 20$ N m^{-2}. The $n = 2$ population, which makes the Balmer lines, rises with T (Boltzmann) and then collapses as H ionises (Saha). Its peak near 9900 K is why Balmer lines are strongest in A0 stars: the spectral sequence is a temperature sequence (Saha 1920; Payne 1925).

Worked example 1.1 — Balmer lines in the Sun and in an A0 star

Compute N_2/N_1 for hydrogen at 5772 K and 9900 K, and the ionisation at 9900 K for $P_e = 20$ Pa ($U_{\rm I} = 2$, $U_{\rm II} = 1$).

Solution.

- 5772 K: $kT = 0.4974$ eV, so $N_2/N_1 = 4e^{-10.20/0.4974} = 5.0 \times 10^{-9}$. Only five atoms in a billion can absorb a Balmer photon; the lines are visible only because H is so abundant.
- 9900 K: $N_2/N_1 = 4e^{-11.96} = 2.6 \times 10^{-5}$, about 5000 times more.
- Saha at 9900 K: $n_e = P_e/kT = 1.46 \times 10^{20}$ m^{-3}; $(2\pi m_e kT/h^2)^{3/2} = 2.38 \times 10^{27}$ m^{-3}; $e^{-13.6/0.8531} = 1.19 \times 10^{-7}$. So $N_{\rm II}/N_{\rm I} = 2.38 \times 10^{27} \times 1.19 \times 10^{-7}/1.46 \times 10^{20} = 1.94$: 66 % ionised.
- Hence $N_2/N_{\rm tot} \approx 2.6 \times 10^{-5}/2.94 = 8.7 \times 10^{-6}$, the peak of Figure 1.3.

Worked example 1.2 — Is Na D in LTE?

Na I D_2: $A_{21} = 6.2 \times 10^7$ s^{-1}, $q_{21} \approx 6 \times 10^{-13}$ m^3 s^{-1}. Compare $n_e \approx 6 \times 10^{19}$ m^{-3} at $\tau_{500} = 1$ with $n_e \approx 10^{17}$ m^{-3} where the line core forms.

Solution. $n_{\rm crit} = A/q \approx 10^{20}$ m^{-3}. At $\tau_{500} = 1$, $C/A \approx 0.6$: close to LTE only with help from photon trapping. Where the core forms, $C/A \approx 10^{-3}$, so $\varepsilon \approx 10^{-3}$: strongly NLTE. The model of Lecture SA3 then gives $S(0)/B \approx 0.14$, and indeed the observed Na D cores are much darker than LTE predicts.

Worked example 1.3 — How close to TE is the Sun's centre?

With photon mean free path $\ell \approx 2$ cm and temperature scale height $H_T \approx R_\odot \approx 7 \times 10^8$ m, estimate the anisotropy $F/(cu)$.

Solution. Diffusion gives $F = \frac{c\ell}{3}|\nabla u|$ with $u \propto T^4$, so $|\nabla u|/u = 4/H_T$ and $F/(cu) = 4\ell/3H_T \approx 4 \times 10^{-11}$. The radiation is Planckian to about one part in 10^{11}, essentially TE, yet this tiny anisotropy carries the whole solar luminosity.

Common misconceptions

1. **"LTE means the radiation is Planckian."** LTE concerns matter; at the surface there is no incoming light, while $B_\nu \neq 0$.
2. **"NLTE means the gas has no temperature."** Electrons and atoms usually still have a Maxwellian kinetic T; only the level populations depart from LTE.
3. **"A Planck spectrum proves TE."** Today's CMB is Planckian but decoupled from matter; free streaming redshifts a Planck spectrum into another Planck spectrum.
4. **"Hydrogen ionises when $kT \approx 13.6$ eV."** It ionises at $kT \approx 0.9$ eV, because of the huge number of free-electron states.

Lecture summary

- The atmosphere is where photons escape; under Collins' five assumptions it is fixed by $T_{\rm eff}$, $\log g$ and composition.

- **TE:** one T, $I_\nu = B_\nu$ (oven, solar core, early Universe). **LTE:** matter at local T, $S_\nu = B_\nu$, $I_\nu \neq B_\nu$ (photospheric continuum, weak lines). **NLTE:** rate equations, $S_\nu \neq B_\nu$ (strong line cores, chromospheres, nebulae, masers).
- Kirchhoff: $j_\nu = \alpha_\nu B_\nu(T)$, valid whenever the matter is in LTE.
- Boltzmann and Saha give the populations; their competition makes the spectral sequence a temperature sequence.

Exam questions and answers — Lecture SA1

Q1. Define TE, LTE and NLTE precisely and give two astrophysical examples of each.

> **A1. TE:** all distributions (Maxwell, Boltzmann, Saha, Planck) at one temperature, with detailed balance and $I_\nu = B_\nu$; examples: the solar core, the pre-recombination Universe. **LTE:** matter in equilibrium at the local temperature, so $S_\nu = B_\nu(T(\mathbf{r}))$, but the radiation may differ from B_ν; examples: the solar photospheric continuum, weak Fe lines in the Sun, Earth's troposphere in the infrared. **NLTE:** some populations differ from Saha–Boltzmann and follow from rate equations coupled to the radiation; examples: Na D and Ca II H & K cores, planetary-nebula forbidden lines, masers.

Q2. Derive Kirchhoff's law and show that it implies $S_\nu = B_\nu$ in LTE. Why can LTE hold although the radiation near a stellar surface is far from a black body?

> **A2.** In a TE cavity, $dI_\nu/ds = 0 = j_\nu - \alpha_\nu B_\nu$, so $j_\nu = \alpha_\nu B_\nu(T)$. Since j_ν and α_ν depend only on the state of the matter, the law holds whenever the matter is in its TE state at the local T; then $S_\nu = B_\nu$. LTE can hold near the surface because particle collisions, which set that state, are frequent (short particle mean free paths), while photons, with long mean free paths, escape. As long as collisional rates dominate radiative ones, the matter does not "notice" the anomalous radiation.

Q3. Explain with Saha and Boltzmann why Balmer lines are strongest in A0 stars ($T \approx 10^4$ K) and weaker in both cooler and hotter stars.

> **A3.** The Balmer opacity is $\propto N_2$. In cool stars $N_2/N_1 \propto e^{-10.2\,\mathrm{eV}/kT}$ is tiny (5×10^{-9} at 5800 K). Above about 10^4 K, Saha ionisation removes neutral H. The product N_2/N_{tot} therefore peaks near 9900 K (for $P_e = 20$ Pa).

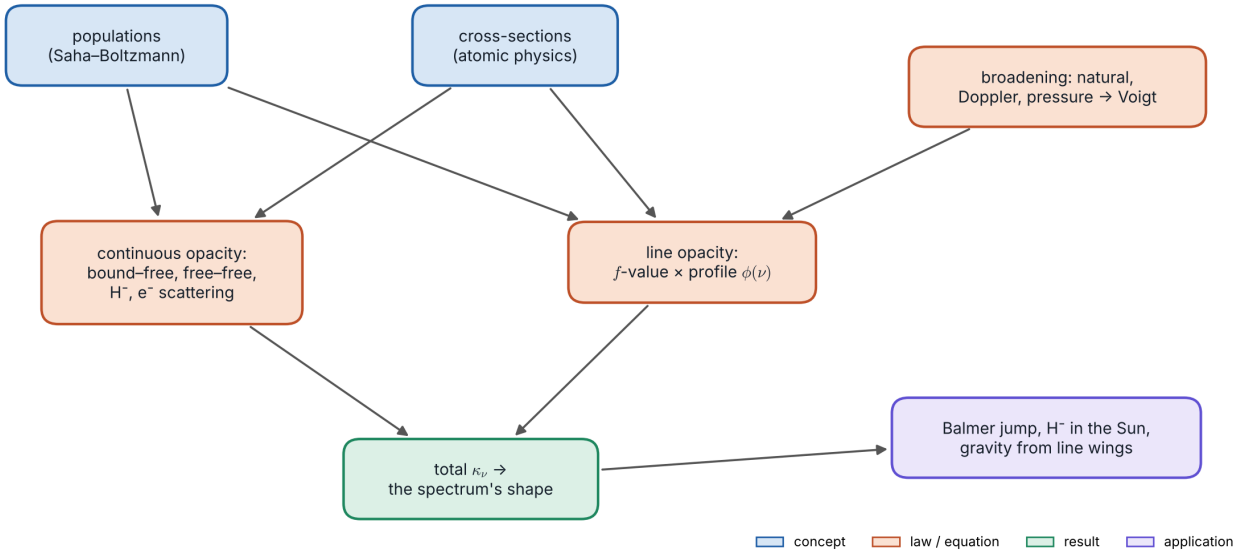
Q4. Today's CMB is an almost perfect Planck spectrum at 2.725 K, yet the photons have not interacted with matter since $z \approx 1100$. Is the CMB in TE? Explain.

> **A4.** It was in TE before recombination, at about 3000 K. Afterwards photons stream freely: I_ν/ν^3 is conserved, and every frequency redshifts by the same factor $1+z$, so a Planck spectrum at T becomes a Planck spectrum at $T/(1+z) \approx 2.7$ K. Today the radiation is **not** in equilibrium with any matter; the Planck shape is a relic. A Planck spectrum alone does not prove present equilibrium.

Lecture SA2 — Where Opacity Comes From: Continuous Absorption and Line Broadening

> **Sources:** Mihalas, *Stellar Atmospheres* (1978) Chs. 4 and 9 (absorption cross-sections; line broadening); Novotny (1973) Ch. 3.4–3.6 (velocities of gas particles, absorption processes, absorption coefficients); Collins §9.2; Rybicki & Lightman Ch. 5 (free–free); Gray, *The Observation and Analysis of Stellar Photospheres*, Chs. 8, 11, 18; Wildt (1939); Barklem, Anstee & O'Mara (2000).

SA2 concept map — where opacity comes from



*Concept map — opacity is **how many absorbers** (populations from Saha–Boltzmann, Lecture SA1) times **how well each absorbs** (atomic cross-sections). Smooth processes give the **continuous opacity**; transitions between bound levels give **lines**, whose width comes from broadening. Their sum shapes the spectrum.*

Learning goals

1. Name the continuous-opacity processes and say which dominates in O, A, G and M stars.
2. Use Kramers' bound–free formula and explain absorption edges and the Balmer jump.
3. Explain why the rare H^- ion controls the Sun's visible opacity.
4. Write the line absorption coefficient, and describe natural, Doppler and pressure broadening and the Voigt profile.
5. Explain why line wings measure gravity and line cores measure temperature.

The big ideas in plain words

Opacity is a menu of processes (Figure 2.1). A photon can lift a bound electron to a higher level (*bound–bound*: a line), free it (*bound–free*), be absorbed by a free electron passing an ion (*free–free*), or be redirected (*scattering*). The last three vary smoothly with frequency and form the **continuous opacity**.

Edges are fingerprints. A photon can ionise an atom from level n only if it has enough energy, so each level absorbs from a sharp threshold (an *edge*) to higher frequencies, fading as ν^{-3}. Hydrogen's edges are at 91.2 nm (Lyman), 364.7 nm (Balmer) and 820.6 nm (Paschen).

Counting absorbers beats counting atoms. A process absorbs only if atoms are in the right state. In the Sun only one H atom in 10^8 is the fragile negative ion H^-, yet it out-absorbs ordinary hydrogen by a factor of 100 in the visible, because almost all neutral hydrogen sits in the ground state, which absorbs only below 91 nm.

Why lines have widths. Three things smear a sharp transition: *uncertainty* (short-lived states have fuzzy energies: natural broadening), *motion* (Doppler shifts: thermal and turbulent broadening) and *neighbours* (passing particles perturb the levels: pressure broadening). The first and third give broad Lorentzian wings; the second gives a Gaussian core. Together: the Voigt profile.

The core is a thermometer; the wings are a barometer. The Doppler core measures temperature and turbulence. The pressure-broadened wings measure the density of perturbers, and through hydrostatic equilibrium the surface gravity.

2.1 The menu of processes

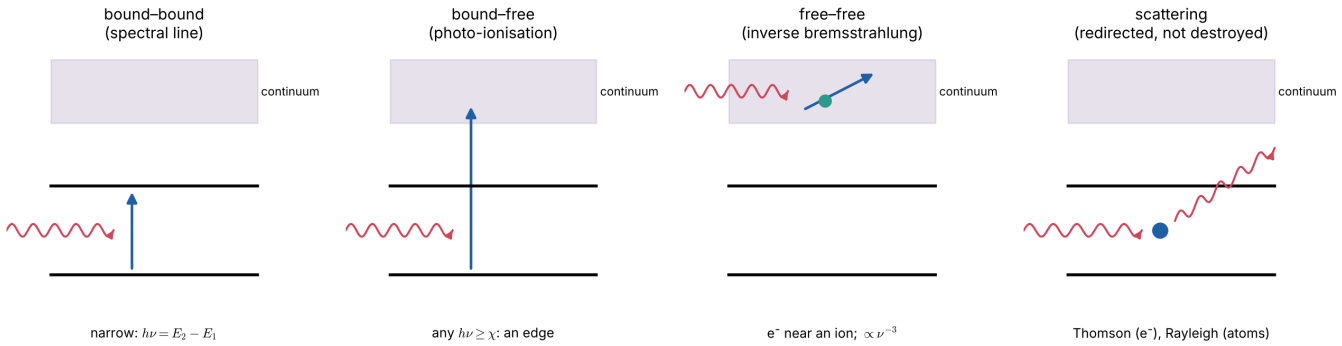


Figure 2.1 — The four ways matter stops a photon, drawn on an energy-level diagram (repeated from Lecture RT2).

Process	Frequency dependence	Dominant in
bound–free	edges, $\propto \nu^{-3}$ above each	H I in A–B stars; He in O–B; metals in the UV; H^- in F–K
free–free	$\propto \nu^{-3}$ optical, ν^{-2} radio	hot stars (IR, radio); H^- free–free beyond 1.6 μm
electron scattering	gray, $\sigma_T = 6.652 \times 10^{-29}$ m^2	O stars, interiors
Rayleigh scattering	$\propto \lambda^{-4}$	blue and UV of cool giants
molecules, dust	bands	M, L, T dwarfs
bound–bound	narrow profiles (§2.4)	lines in all stars

2.2 Hydrogenic bound–free and free–free absorption

Kramers' cross-section for photo-ionisation from level n of a hydrogen-like ion of charge Z:

$$\boxed{\sigma_{\rm bf}(n,\nu) = 2.815\times10^{29}\,\frac{Z^4\,g_{\rm bf}}{n^5\nu^3}\ \mathrm{cm}^2 \qquad (\nu \ge \nu_n = 3.288\times10^{15}\,Z^2/n^2\ \mathrm{Hz})}$$

At its own edge the cross-section *grows* as n, so excited levels are excellent absorbers **if they are populated**. In LTE, with Boltzmann populations and stimulated emission:

$$\alpha_\nu^{\rm bf} = \sum_n N_n\,\sigma_{\rm bf}(n,\nu)\left(1-e^{-h\nu/kT}\right), \qquad N_n = N_{\rm H\,I}\frac{2n^2 e^{-E_n/kT}}{U(T)}.$$

Free–free (inverse bremsstrahlung) for a hydrogen-like ion: $\alpha_\nu^{\rm ff} = 3.69\times10^{8}\,Z^2 n_e n_i T^{-1/2}\nu^{-3} g_{\rm ff}(1-e^{-h\nu/kT})\ \mathrm{cm}^{-1}$ (cgs). It rises toward low frequency and dominates the radio emission of H II regions (Lecture RT3).

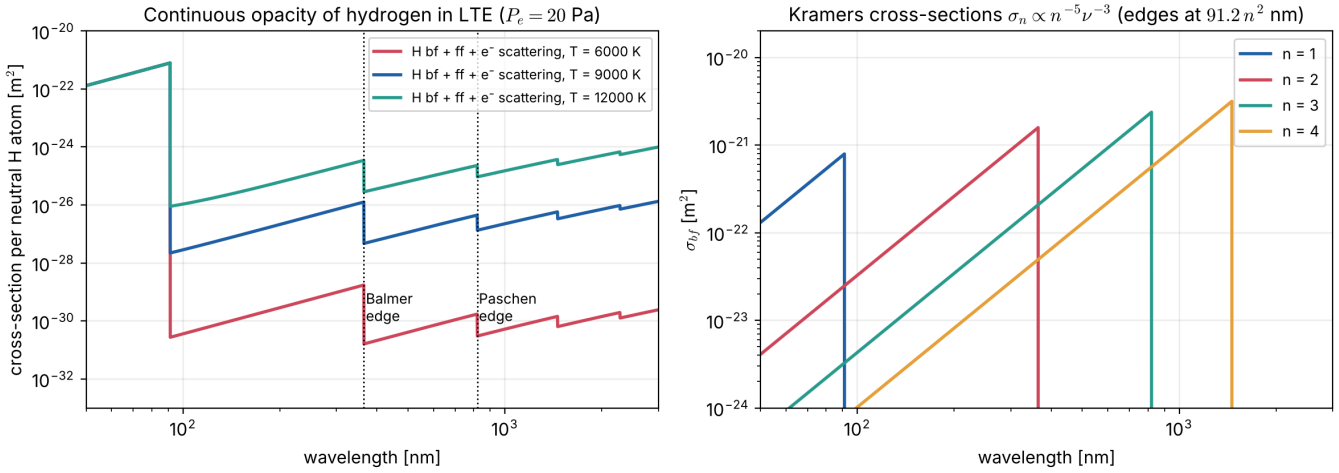


Figure 2.2 — Left: hydrogen continuous cross-section per neutral H atom in LTE at $P_e = 20$ Pa (`scripts/sa_physics.py`). The Lyman, Balmer and Paschen jumps grow dramatically with T, because excited-level populations follow Boltzmann. Right: Kramers cross-sections of the first four levels, each absorbing only blueward of its edge.

The Balmer jump across the spectral sequence. At $P_e = 20$ Pa the hydrogen opacity jumps by factors of 106, 26 and 12 across the Balmer edge at 6000, 9000 and 12 000 K. Yet the *observed* jump is largest in late B and A stars. In the Sun, H^- (with no edge at 364.7 nm) dominates and dilutes the jump. In A stars hydrogen dominates the continuum: blueward of the edge we see higher, cooler layers (Eddington–Barbier), and the flux drops. In hotter stars ionisation and electron scattering shrink it. The jump measures T in B stars and gravity (through P_e) in A–F stars.

2.3 The negative hydrogen ion

- Binding energy 0.754 eV, so bound–free absorption extends to 1.64 µm, with a cross-section peaking near 850 nm at $\approx 4\times10^{-21}$ m^2. Free–free H^- dominates beyond 1.6 µm. The opacity minimum at 1.6 µm lets us see deepest there.
- Abundance from a Saha equation with the electron *added*:

$$\frac{n(\mathrm{H}^-)}{n(\mathrm{H\,I})\,n_e} = \frac{1}{4}\left(\frac{h^2}{2\pi m_e kT}\right)^{3/2} e^{+0.754\,\mathrm{eV}/kT}.$$

- **The electrons come from metals** (Na, Mg, Al, Si, Ca, Fe, with ionisation energies of 5–8 eV), because hydrogen is almost neutral. So H^- opacity scales with metallicity: metal-poor stars are more transparent, and we see deeper, hotter layers.
- Identified by Wildt (1939); cross-sections computed by Chandrasekhar in the 1940s.

Total and mean opacities. $\kappa_\nu \rho = \sum(\alpha^{\rm bf} + \alpha^{\rm ff})(1 - e^{-h\nu/kT}) + \alpha^{\rm es} + \alpha^{\rm Ray} + \alpha^{\rm lines}$; stimulated emission applies to true absorption only. Structure calculations use the Rosseland mean (for flux, Lecture RT4) and the Planck mean, taken from tables (OPAL, Opacity Project, Ferguson et al. 2005). Deep inside stars, bound–free and free–free combine into **Kramers' law**, $\bar\kappa \propto \rho T^{-3.5}$.

2.4 Line absorption

For $l \to u$ with oscillator strength f_{lu} and normalised profile $\phi(\nu)$, in LTE (cgs):

$$\boxed{\alpha_\nu^{\rm line} = \frac{\pi e^2}{m_e c} f_{lu}\, n_l\, \phi(\nu)\big(1 - e^{-h\nu/kT}\big)}, \qquad \frac{\pi e^2}{m_e c} = 0.02654 \ \mathrm{cm^2\,Hz}.$$

Line strengths are quoted as $\log gf$, and $A_{ul} = 0.6670\,(g_l/g_u)\,f_{lu}/\lambda^2\ \mathrm{s^{-1}}$ (λ in cm).

Mechanism	Physical cause	Profile	Width
natural	finite lifetimes	Lorentzian	$\gamma_{\rm rad} = \Gamma_u + \Gamma_l$; $\Delta\lambda = \lambda^2\gamma/2\pi c$
Doppler	thermal motion + microturbulence ξ	Gaussian	$\Delta\nu_D = (\nu_0/c)\sqrt{2kT/m + \xi^2}$
pressure, $\Delta\nu = C_n/r^n$	perturbing particles	Lorentzian (impact) or Holtsmark (quasi-static)	$\gamma_n \propto N_{\rm pert}$

Pressure-broadening families: $n = 2$ linear Stark (H and He II lines in hot stars and white dwarfs); $n = 3$ resonance (self) broadening; $n = 4$ quadratic Stark (non-hydrogenic ions in hot stars); $n = 6$ van der Waals by neutral H (metal lines in cool stars, e.g. the Na D and Mg I b wings). Modern van der Waals widths use the **ABO theory**, which corrects Unsöld's classical values by factors of 1.5–2.5.

The Voigt profile combines them:

$$\phi(\nu) = \frac{H(a,v)}{\sqrt{\pi}\,\Delta\nu_D}, \qquad H(a,v) = \frac{a}{\pi}\int \frac{e^{-y^2}\,dy}{(v-y)^2 + a^2}, \qquad a = \frac{\gamma}{4\pi\Delta\nu_D},\ v = \frac{\nu - \nu_0}{\Delta\nu_D}.$$

The core is $H \approx e^{-v^2}$ (Gaussian); the wings are $H \approx a/(\sqrt{\pi}v^2)$ (Lorentzian), taking over beyond $|v| \approx 2$–3 for $a \sim 0.01$.

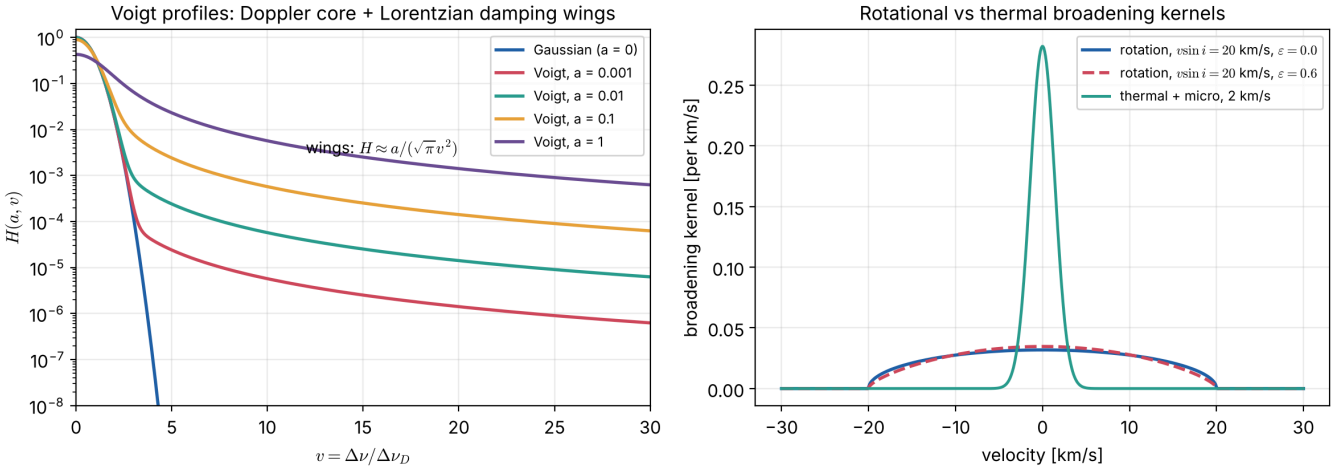


Figure 2.3 — Left: Voigt functions for increasing damping a; Lorentzian wings rise above the Gaussian core beyond $v \approx 2$–3. Right: the rotational broadening kernel for $v \sin i = 20$ km s^{-1}, without and with limb darkening, compared with a 2 km s^{-1} thermal Gaussian.

Broadening of the emergent spectrum. Rotation ($\pm v \sin i$, with the kernel of Gray Ch. 18), macroturbulence, the instrument ($R = \lambda/\Delta\lambda$) and Zeeman splitting (Lecture RT5) act on the *emergent* spectrum. They change the shape but not the equivalent width. Atomic broadening acts on the *absorption coefficient*, and for saturated lines it does change the equivalent width (Lecture SA4).

Worked example 2.1 — Why H$^-$ dominates the solar opacity at 500 nm

At $T = 6000$ K and $n_e = 10^{19}$ m^{-3}, compare H$^-$ bound–free with hydrogen $n = 3$ (Paschen) bound–free, per H atom.

Solution.

1. $(2\pi m_e kT/h^2)^{3/2} = 1.12 \times 10^{27}$ m^{-3} and $e^{-0.754/0.517} = 0.233$, so $n(\mathrm{H^-})/n(\mathrm{H\,I}) = 10^{19}/(4 \times 1.12 \times 10^{27} \times 0.233) = 9.6 \times 10^{-9}$. Times 4×10^{-21} m^2 gives 3.8×10^{-29} m^2 per H atom.
2. $N_3/N_{\mathrm{H\,I}} = 9e^{-12.09/0.517} = 6.3 \times 10^{-10}$, and $\sigma_3(500\,\mathrm{nm}) = 5.4 \times 10^{-22}$ m^2, giving 3.4×10^{-31} m^2 per H atom.
3. The ratio is ≈110: a tiny fraction of fragile ions out-absorbs excited hydrogen by two orders of magnitude.

Worked example 2.2 — How narrow is natural broadening?

For Hα the upper level decays at 1.0×10^8 s^{-1} and the lower level (by Lyα) at 6.27×10^8 s^{-1}. Compare the natural FWHM with the Doppler width at 5800 K (0.21 Å).

Solution. $\gamma = 7.3 \times 10^8$ s^{-1}, so $\Delta\lambda = \lambda^2\gamma/2\pi c = 1.7 \times 10^{-13}$ m $= 0.0017$ Å, about 120 times narrower than the Doppler core. Natural broadening shows only in the far wings.

Worked example 2.3 — The damping parameter of a solar Fe I line

An Fe I line at 500 nm has $\gamma \approx 10^9$ s^{-1} (mostly van der Waals); $T = 5772$ K and $\xi = 1$ km s^{-1}. Find a.

Solution. $v_D = \sqrt{2kT/m_{\mathrm{Fe}} + \xi^2} = 1.65$ km s^{-1}, so $\Delta\nu_D = v_D/\lambda = 3.3 \times 10^9$ Hz and $a = \gamma/(4\pi\Delta\nu_D) = 0.024$. That is why $a \approx 0.01$–0.03 is used for solar metal lines in the curve of growth (Lecture SA4).

Common misconceptions

1. **"The most abundant atom gives the most opacity."** What counts is the number of particles in an absorbing state at that frequency.
2. **"Bound–free absorption occurs only at the edge."** It occurs at all frequencies *above* the edge, falling roughly as ν^{-3}.
3. **"The natural width is what we see."** It is usually 10–1000 times smaller than the Doppler and pressure widths.
4. **"The Voigt core is Lorentzian."** The core is essentially Gaussian; the Lorentzian shows only in the wings.

Lecture summary

- Opacity = number of absorbers (Saha–Boltzmann) × cross-section, summed over bound–free, free–free, scattering and lines.
- Kramers: $\sigma_{\mathrm{bf}} \propto Z^4/(n^5\nu^3)$ above each edge; edges make the Balmer and Paschen jumps.
- H$^-$ (0.754 eV, to 1.64 μm) dominates F–K photospheres; its abundance tracks the metal-supplied electrons.
- Lines: $\alpha^{\mathrm{line}} = (\pi e^2/m_e c) f\, n_l \phi(\nu)(1 - e^{-h\nu/kT})$, with a Voigt profile: Doppler core (temperature) and pressure wings (gravity).
- Rotation, macroturbulence and the instrument broaden the emergent spectrum without changing equivalent widths.

Exam questions and answers — Lecture SA2

Q1. Explain why H$^-$ is the main continuous opacity in the solar photosphere, and why metal-poor stars are more transparent.

> **A1.** Solar hydrogen is mostly neutral and in the ground state, which absorbs only below 91 nm; excited levels are barely populated ($N_3/N_1 \sim 10^{-9}$). H$^-$, though only $\sim 10^{-8}$ of H, is ionised by any photon below 1.64 μm with a large cross-section, so it dominates the visible and near-IR. Its abundance is $\propto n_e$, and the electrons come from metals: fewer metals means fewer electrons, less H$^-$ and lower opacity, so we see deeper layers.

Q2. Write Kramers' bound–free cross-section, show that at its own edge it is proportional to n, and rank the dominant continuous opacity in an O star, an A0 star, the Sun and an M dwarf.

> **A2.** $\sigma_{\mathrm{bf}} = 2.815 \times 10^{29} Z^4 g/(n^5\nu^3)$ cm^2 for $\nu \geq \nu_1/n^2$. At the edge $\nu_n^3 = \nu_1^3/n^6$, so $\sigma \propto n^6/n^5 = n$. O star: electron scattering (plus He bound–free). A0: H I bound–free and free–free. Sun: H$^-$. M dwarf: H$^-$ plus molecules.

Q3. Name the three atomic broadening mechanisms and the profile each gives. Which part of a strong line does each control?

> **A3.** Natural (finite lifetimes) gives a Lorentzian. Pressure (perturbers) gives a Lorentzian in the impact limit, or Holtsmark in the quasi-static limit. Doppler (thermal and turbulent motions) gives a Gaussian. Their convolution is the Voigt profile. The Doppler part controls the core (temperature, microturbulence); the Lorentzian parts control the far wings (pressure, hence gravity).

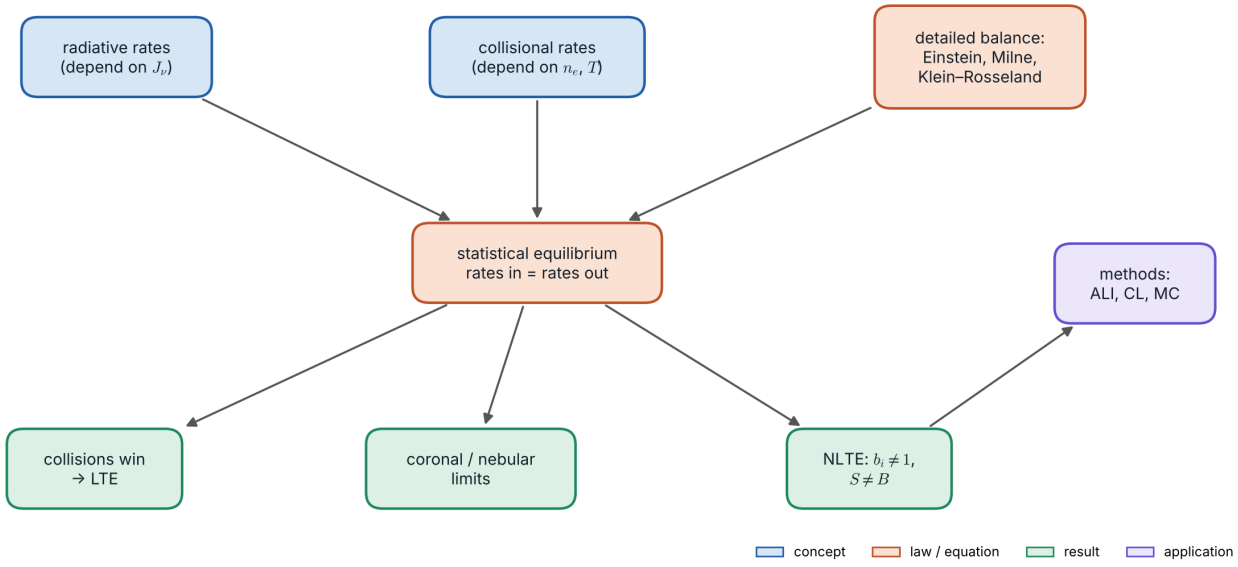
Q4. Why do the wings of strong lines measure surface gravity? Give the chain of reasoning.

> **A4.** Wing strength $\propto \gamma_{\text{pressure}} \propto$ the number density of perturbers where the wings form. By hydrostatic equilibrium the pressure at a given optical depth scales as $P \approx g\tau/\kappa$, so higher g means higher density, more damping, and broader wings at fixed T_{eff} and abundance. Fitting the wings of Balmer lines (hot stars) or Mg I b and Ca II (cool stars) therefore gives $\log g$.

Lecture SA3 — Transition Rates, Statistical Equilibrium and Non-LTE

> **Sources:** Mihalas, *Stellar Atmospheres* (1978) Chs. 5, 11–12 (rates, statistical equilibrium, NLTE line formation); Collins §9.1; R. N. Thomas (1965); Hubeny & Mihalas, *Theory of Stellar Atmospheres* (2014); Osterbrock & Ferland, *Astrophysics of Gaseous Nebulae* (2006); van Regemorter (1962); Drawin (1968).

SA3 concept map — rates and non-LTE



*Concept map — populations are set by a **balance of traffic**: radiative rates (which depend on the light J_ν) and collisional rates (which depend on n_e and T), each tied to its inverse by detailed balance. When collisions win, the result is LTE. In thin gas we reach the coronal and nebular limits. In between lies the full NLTE problem, with its own solution methods.*

Learning goals

1. List the radiative and collisional processes and the detailed-balance relation linking each to its inverse.
2. Write the statistical-equilibrium equations and say when they reduce to LTE.
3. Derive the two-level-atom source function $S = (1-\varepsilon)\bar{J} + \varepsilon B$ and the critical density.
4. Use the coronal and nebular limits.
5. Explain the main NLTE mechanisms with departure coefficients, and name the solution methods and codes.

The big ideas in plain words

A population is traffic in and traffic out. Picture each atomic level as a room with doors to every other room and to the "continuum" (ion plus free electron). Atoms leave a room by absorbing, emitting or being hit by an electron, and enter by the inverse processes. In a steady state, traffic in equals traffic out for every room: **statistical equilibrium**.

Two kinds of traffic. *Radiative* transitions depend on the light, which may come from far away at a different temperature, or escape. *Collisional* transitions depend on the local electron density and temperature. Collisions push towards LTE; radiation pushes towards whatever the light dictates.

Every process has a partner. The probability of a process and of its inverse are linked by atomic physics alone. Imagine the gas in full TE, where each process balances its inverse, read off the link, and it then holds everywhere. Einstein (bound–bound), Milne (bound–free) and Klein–Rosseland (collisions) are three versions of one argument.

NLTE is the rule in thin gas. Most of the Universe by volume, including nebulae, the ISM, chromospheres, coronae and winds, is NLTE. Assuming LTE there misreads line strengths and gives wrong abundances, sometimes by a factor of two or more.

3.1 Inventory of rates and detailed balance

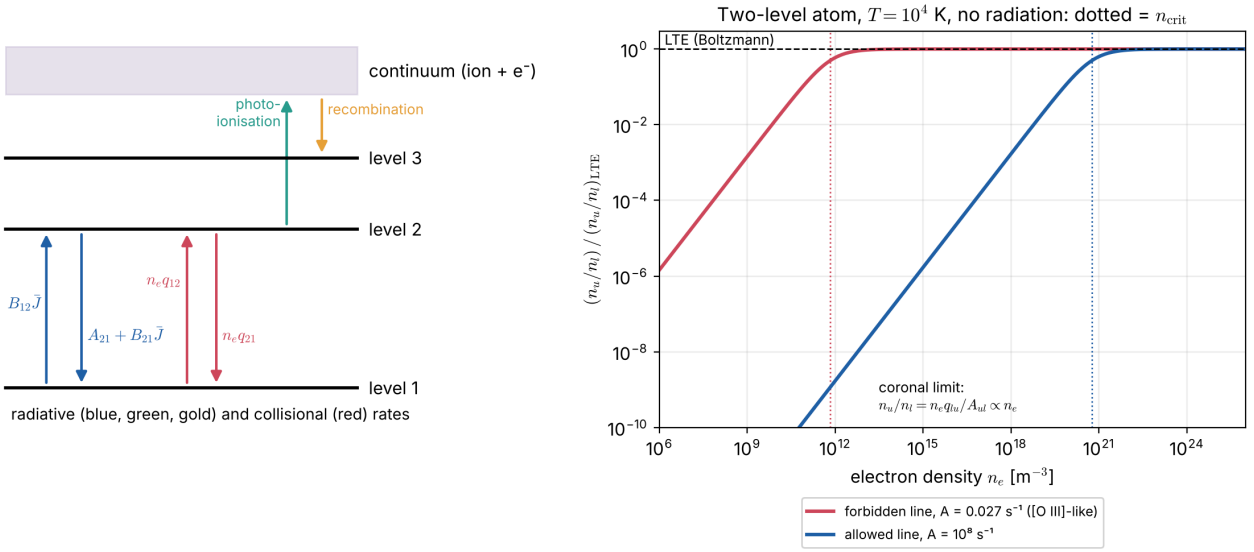


Figure 3.1 — Left: radiative and collisional processes between bound levels and the continuum. Right: for an optically thin two-level atom with no radiation ($T = 10^4$ K), the upper population relative to LTE rises $\propto n_e$ (coronal regime) and saturates above the critical density. A forbidden line ($A = 0.027\ s^{-1}$) reaches LTE above $\sim 7 \times 10^{11}\ m^{-3}$; an allowed line ($A = 10^8\ s^{-1}$) needs $\sim 6 \times 10^{20}\ m^{-3}$.

Process (rate per atom)	Inverse	Link from detailed balance
radiative excitation $B_{lu}\bar{J}$	emission $A_{ul} + B_{ul}\bar{J}$	**Einstein:** $g_l B_{lu} = g_u B_{ul}$, $A_{ul} = \frac{2h\nu^3}{c^2} B_{ul}$
photo-ionisation $4\pi \int \frac{\sigma_i}{h\nu} J_\nu d\nu$	radiative recombination	**Milne** relation
collisional excitation $n_e q_{lu}(T)$	de-excitation $n_e q_{ul}$	**Klein–Rosseland:** $\frac{q_{lu}}{q_{ul}} = \frac{g_u}{g_l} e^{-\Delta E/kT}$
collisional ionisation $n_e q_{i\kappa}$	three-body recombination ($\propto n_e^2$)	**Saha**
dielectronic recombination; charge exchange	—	important in hot plasmas and nebulae

Collision strengths. Rate coefficients are written with the dimensionless, symmetric collision strength Ω_{lu}:

$$q_{ul} = \frac{8.629 \times 10^{-6}}{g_u \sqrt{T}}\,\Omega_{lu}\ \mathrm{cm^3\,s^{-1}} \quad (\times 10^{-6}\ \text{in}\ \mathrm{m^3\,s^{-1}}), \qquad q_{lu} = \frac{g_u}{g_l} q_{ul} e^{-\Delta E/kT}.$$

Without data, approximations are used: van Regemorter (allowed lines), Seaton (ionisation), and Drawin (neutral-H collisions in cool stars, now replaced by quantum calculations).

3.2 Statistical equilibrium

For every level i:

$$\boxed{n_i \sum_{j\neq i} (R_{ij} + C_{ij}) = \sum_{j\neq i} n_j (R_{ji} + C_{ji})} \qquad (\text{rates out} = \text{rates in})$$

together with particle and charge conservation. If J_ν were known, this would be a linear system $\mathbf{An} = \mathbf{b}$. But J_ν depends on the populations *everywhere* through the transfer equation, so the full problem is **non-local and non-linear**.

LTE is a limit, reached in two ways: collisions dominate every transition ($C \gg R$), or the radiation is Planckian ($J_\nu = B_\nu$, so radiative rates balance pairwise). Either way, Saha–Boltzmann follows.

The two-level atom

Statistical equilibrium for $1 \leftrightarrow 2$, $n_1(B_{12}\bar{J} + C_{12}) = n_2(A_{21} + B_{21}\bar{J} + C_{21})$, inserted into the line source function of Lecture RT2, gives

$$\boxed{S^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B_\nu(T), \qquad \varepsilon = \frac{\varepsilon'}{1+\varepsilon'}, \quad \varepsilon' = \frac{C_{21}}{A_{21}}\left(1 - e^{-h\nu/kT}\right)}$$

This is exactly the scattering source function of Lecture RT4, now derived from atomic physics: ε is the probability that a photon absorbed in the line is destroyed by a collision rather than re-emitted.

- **Collisions dominate** ($n_e \gg n_{\text{crit}} = A_{21}/q_{21}$): $\varepsilon \to 1$, $S \to B$, LTE.
- **Radiation dominates:** $\varepsilon \ll 1$, $S \approx \bar{J}$, scattering-like, NLTE. Near the surface $S(0) \approx \sqrt{\varepsilon} B$.
- **Photon trapping** helps: an optically thick line re-absorbs its own photons, reducing the effective A by the escape probability $\beta_{\text{esc}} \approx 1/\tau$. The practical criterion is therefore $n_e \gg A\,\beta_{\text{esc}}/q$.

3.3 Two classical limits

Coronal approximation (hot, very thin, optically thin, no strong radiation; e.g. the solar corona, supernova remnants). Every collisional excitation produces one escaping photon: $n_l n_e q_{lu} = n_u A_{ul}$, so $n_u/n_l = n_e q_{lu}/A_{ul}$, far below Boltzmann. Ionisation and recombination are both $\propto n_e$, so $n_{i+1}/n_i = q_{\text{ion}}(T)/\alpha_{\text{rec}}(T)$ depends on **temperature only**, which makes Fe IX–XVI lines coronal thermometers.

Nebular approximation (photo-ionised gas around a hot star). Diluted starlight, $J_\nu \approx W B_\nu(T_*)$ with $W \approx \frac{1}{4}(R_*/r)^2 \ll 1$, ionises from the ground state; recombination cascades populate the excited levels; forbidden lines are collisionally excited with $n_e \ll n_{\text{crit}}$.

3.4 Non-LTE in stellar atmospheres

Departure coefficients $b_i = n_i/n_i^{\mathrm{LTE}}$ (= 1 in LTE) give the line source function

$$S^{\mathrm{line}} = \frac{2h\nu^3}{c^2}\frac{1}{\frac{b_l}{b_u}e^{h\nu/kT}-1} \approx \frac{b_u}{b_l}B_\nu(T) \quad \text{(Wien limit)}.$$

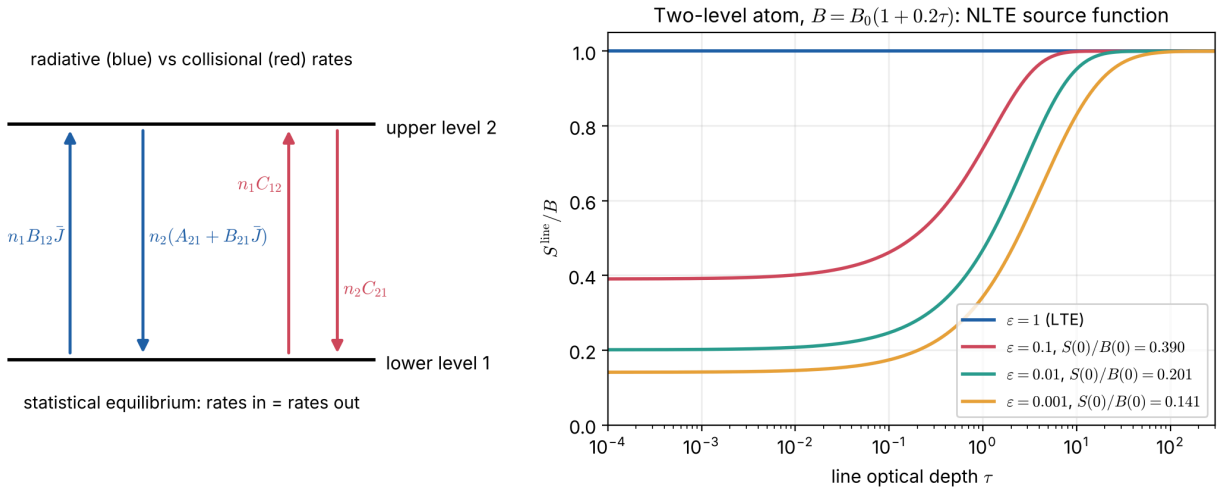


Figure 3.2 — Left: the four rate processes of a two-level atom. Right: the exact line source function (Λ-matrix solution, `scripts/lte_solvers.py`) in an atmosphere with B rising gently inward. As collisions become rarer ($\varepsilon\downarrow$), the surface S falls far below B, and thermalisation happens deeper, at $\tau\sim 1/\sqrt{\varepsilon}$.

Mechanism	What happens	Effect on the spectrum	Typical cases
photon losses	line photons escape near the surface: $\bar{J}<B$, $b_u<b_l$, $S<B$	line cores darker than LTE	Na D, Ca II H & K, Hα core, Mg II h & k
overionisation	UV $J_\nu > B_\nu$ ionises minority species	weaker lines; LTE underestimates abundances	Fe I, Mg I in metal-poor stars
overrecombination	recombination cascades overpopulate upper levels	emission features	He I, H I in nebulae and winds
pumping, fluorescence	strong radiation fills particular levels	emission, altered strengths	Bowen fluorescence (O III), masers
photon suction	cascades through strong lines drain a level	modified populations	high-excitation solar lines

NLTE abundance corrections are about −0.2 dex for the O I 777 nm triplet in the Sun, and up to a few tenths of a dex for Fe I and Mg I in metal-poor giants.

Solution methods (names only).

- *Formal solvers:* Feautrier; long and short characteristics (linear, parabolic, Bézier); DELO and Hermitian methods; discontinuous finite elements; discrete ordinates; Monte Carlo.
- *Iteration schemes:* Λ-iteration; core saturation; accelerated Λ-iteration (ALI) and multilevel ALI (MALI); Gauss–Seidel and SOR; multigrid; Ng acceleration; Krylov methods (GMRES, BiCGStab); complete linearization; hybrid CL/ALI; variable Eddington factors; the equivalent two-level atom; escape-probability and Sobolev methods; comoving-frame and partial-redistribution (PRD) schemes.

- *Codes:* MULTI, MULTI3D, RH, Lightweaver, DETAIL/SURFACE, TLUSTY/SYNSPEC, PHOENIX, CMFGEN, FASTWIND, PoWR, Balder, Cloudy, MOCASSIN, LIME, RADEX.

Worked example 3.1 — [O III] in a planetary nebula

The $^1D_2$ *level of* O^{2+} ($g_u = 5$) *has* $A \approx 0.027\ s^{-1}$ *and* $\Omega \approx 2.29$. *At* $T = 10^4$ *K find* q_{ul} *and* $n_{\rm crit}$, *and the regime in a nebula with* $n_e = 10^{10}\ m^{-3}$.

Solution. $q_{ul} = 8.629 \times 10^{-12} \times 2.29/(5 \times 100) = 3.95 \times 10^{-14}$ m^3 s^{-1}, so $n_{\rm crit} = A/q_{ul} = 6.8 \times 10^{11}$ m^{-3}. In the nebula $n_e/n_{\rm crit} \approx 0.015$, and the upper level holds only 1.4 % of its LTE population (Figure 3.1): extreme NLTE. Every excitation yields a photon, so the emissivity is $\propto n_e n_{\rm ion} q_{lu}(T_e)$. The ratio [O III] 436.3/500.7 nm then depends only on T_e, which is how nebular temperatures (~10 000 K) are measured.

Worked example 3.2 — An allowed line from corona to photosphere

Allowed line: $A = 10^8\ s^{-1}$, $\Omega = 10$, $g_l = g_u = 5$, $\Delta E = 2.48$ *eV*, $T = 10^4$ *K. Find* $n_{\rm crit}$ *and the population relative to LTE at* $n_e = 10^{15}$ *and* $10^{20}\ m^{-3}$.

Solution. $q_{ul} = 8.629 \times 10^{-12} \times 10/500 = 1.73 \times 10^{-13}$ m^3 s^{-1}, so $n_{\rm crit} = 5.8 \times 10^{20}$ m^{-3}. Below it the ratio is $\approx n_e/n_{\rm crit}$: 1.7×10^{-6} in the transition region (coronal) and 0.15 in the deep photosphere. Even photospheres cannot thermalise strong allowed lines by collisions alone; photon trapping must help.

Worked example 3.3 — Klein–Rosseland from detailed balance

Derive the relation between q_{lu} *and* q_{ul}.

Solution. In TE, collisional excitation balances de-excitation: $n_l n_e q_{lu} = n_u n_e q_{ul}$, with Boltzmann populations. So $q_{lu}/q_{ul} = (g_u/g_l)e^{-\Delta E/kT}$. The coefficients depend only on cross-sections and on the Maxwellian at T, so the relation holds in any gas with Maxwellian electrons, however far from TE.

Common misconceptions

1. **“Detailed-balance relations apply only in TE.”** They are *derived* in TE but involve only atomic properties, so they hold everywhere.
2. **“High temperature means LTE.”** The corona is very hot and far from LTE, because its density is tiny. What matters is collisional versus radiative rates.
3. **“NLTE always weakens lines.”** Photon losses deepen cores, overionisation weakens minority-species lines, and pumping can create emission.
4. **“Dense atmospheres are LTE everywhere.”** Near the surface photons escape ($\beta_{\rm esc} \to 1$), so even dense atmospheres show NLTE in strong lines.

Lecture summary

- Radiative rates depend on J_ν; collisional rates on n_e and T. Detailed balance links each to its inverse (Einstein, Milne, Klein–Rosseland, Saha).

- Statistical equilibrium, $n_i \sum P_{ij} = \sum n_j P_{ji}$, coupled to the transfer equation, is non-local and non-linear. LTE results when collisions dominate or $J = B$.
- Two-level atom: $S = (1-\varepsilon)\bar{J} + \varepsilon B$, with $n_{\rm crit} = A/q$; photon trapping lowers the effective $n_{\rm crit}$.
- Coronal limit: $n_u/n_l = n_e q_{lu}/A$, and ionisation depends on T only. Nebular limit: diluted photo-ionisation versus recombination.
- NLTE mechanisms: photon losses, overionisation, overrecombination, pumping. Methods: ALI/MALI, complete linearization, CL/ALI and others.

Exam questions and answers — Lecture SA3

Q1. List the radiative and collisional processes that populate and empty a bound level, and give the detailed-balance relation for each pair. Why do these relations hold outside TE?

> **A1.** Radiative excitation $B_{lu}\bar{J}$ ↔ emission $A_{ul} + B_{ul}\bar{J}$ (Einstein). Photo-ionisation ↔ radiative recombination (Milne). Collisional excitation $n_e q_{lu}$ ↔ de-excitation $n_e q_{ul}$ (Klein–Rosseland, $q_{lu}/q_{ul} = (g_u/g_l)e^{-\Delta E/kT}$). Collisional ionisation ↔ three-body recombination (Saha). They involve only atomic cross-sections and, for collisions, the Maxwellian electron distribution, so they hold whatever the radiation field or populations.

Q2. For a two-level atom, write $S^{\rm line}$ in terms of $\bar{J}$, B and ε. When does the line form in LTE, and why can a dense photosphere still show NLTE line cores?

> **A2.** $S = (1-\varepsilon)\bar{J} + \varepsilon B$ with $\varepsilon'/(1+\varepsilon') = \varepsilon$ and $\varepsilon' = (C_{21}/A_{21})(1 - e^{-h\nu/kT})$. LTE requires $C_{21} \gg A_{21}$, i.e. $n_e \gg n_{\rm crit}$ (helped by trapping, $n_e \gg A\beta_{\rm esc}/q$), **and** depths below $\tau \sim 1/\sqrt{\varepsilon}$. Line cores form high, where n_e is low and photons escape ($\beta_{\rm esc} \to 1$), so $\varepsilon \ll 1$ and $S < B$ there.

Q3. Describe the coronal approximation, show that its ionisation balance is independent of density, and give one application.

> **A3.** Thin gas, weak radiation: collisional excitation is balanced by spontaneous decay ($n_l n_e q_{lu} = n_u A_{ul}$), and collisional ionisation by radiative plus dielectronic recombination. Both rates per ion are $\propto n_e$, so $n_{i+1}/n_i = q_{\rm ion}(T)/\alpha_{\rm rec}(T)$. Application: Fe IX–XVI ion ratios measure coronal temperatures of 1–3 MK.

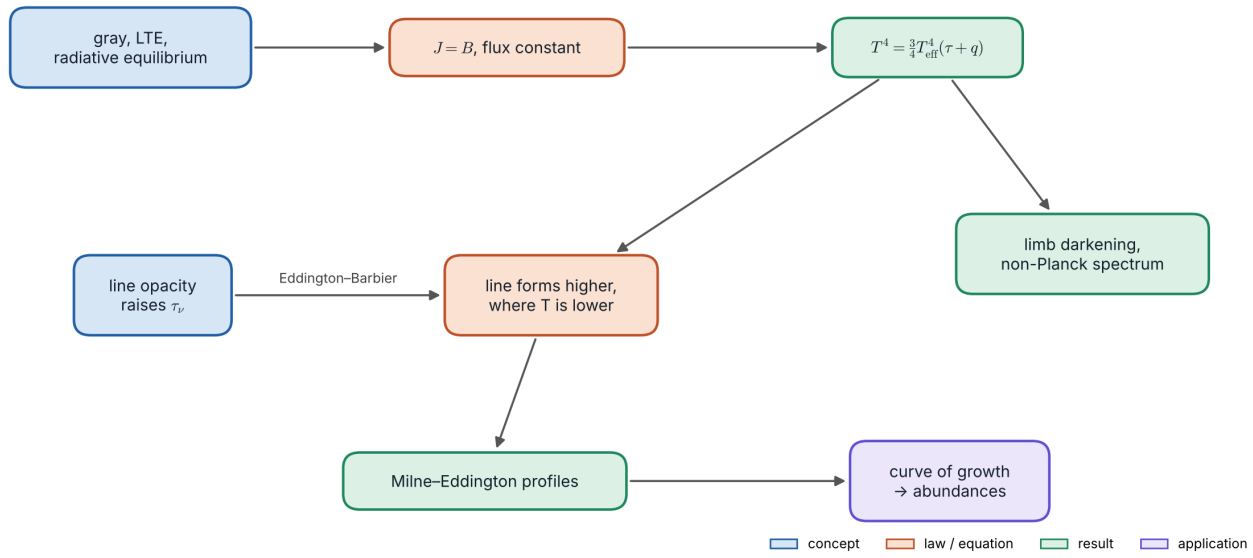
Q4. Why are strong resonance-line cores darker in NLTE than in LTE? Use departure coefficients, and name two methods and two codes used to compute them.

> **A4.** Near the surface line photons escape, so $\bar{J} < B$ and fewer atoms are radiatively excited: $b_u < b_l$, so $S \approx (b_u/b_l)B < B$. By Eddington–Barbier the core intensity is $\approx S$ at $\tau_\nu \approx 1$, below the LTE value. Methods: e.g. ALI/MALI, complete linearization, short characteristics, Feautrier. Codes: e.g. MULTI, RH, TLUSTY, PHOENIX, CMFGEN, DETAIL.

Lecture SA4 — The Gray LTE Atmosphere and How Spectral Lines Form

> **Sources:** Collins §9.4 (gray atmosphere, Eddington approximation); Chandrasekhar (1950) Chs. III–V (Milne's equation, Hopf function, H-functions); Novotny (1973) Ch. 2.4–2.5 and Ch. 5 (formation of absorption lines, line profiles, line strengths); Mihalas (1978) Chs. 3 and 10; Gray, *The Observation and Analysis of Stellar Photospheres*, Chs. 13–16.

SA4 concept map — the gray atmosphere and LTE lines



*Concept map — the top row solves the simplest complete atmosphere: **gray, LTE, radiative equilibrium**, giving $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + q)$, limb darkening and a non-Planckian spectrum. The bottom row adds **line opacity**: it raises τ_ν, so lines are seen higher, where it is cooler (Eddington–Barbier). Milne–Eddington profiles and the curve of growth then turn line strengths into abundances.*

Learning goals

1. Derive $T^4 = \frac{3}{4}T_{\rm eff}^4\left(\tau + \frac{2}{3}\right)$ step by step, and know where the approximation enters.
2. State the exact (Hopf) result and the gray limb-darkening laws.
3. Explain why a gray atmosphere still emits a non-Planckian spectrum.
4. Explain line formation in LTE with Eddington–Barbier, and why an isothermal atmosphere shows no lines.
5. Derive the Milne–Eddington profile and describe the three parts of the curve of growth.

The big ideas in plain words

The gray atmosphere is the hydrogen atom of stellar atmospheres. Make the opacity the same at all frequencies, assume LTE and radiative equilibrium, and the whole problem can be solved exactly. It gives the temperature structure, limb darkening and the spectrum, and it is the starting point of every real model.

In radiative equilibrium the gas absorbs exactly what it emits. With gray opacity this means $J = B$: the mean intensity equals the Planck function at every depth, while the constant flux $\sigma T_{\rm eff}^4$ flows outward through the layers.

The "surface temperature" is the temperature at $\tau = 2/3$. Effective temperature is not the temperature at the top; it is the temperature of the layer from which the flux appears to come. The top layers are cooler, about $0.81\,T_{\rm eff}$.

Lines are windows to higher layers (Figure 4.3). Extra opacity at line frequencies means we see only the upper, cooler gas there. Lower T means lower B_ν, which means a darker line. No temperature gradient, no lines; a temperature *rise* outward gives emission lines.

A line has three lives as it grows. While weak, every added atom deepens it (linear). Once the core is black, adding atoms only widens it slowly (saturated, flat). Finally the far Lorentzian wings become opaque, and the line grows again (damping).

4.1 The gray atmosphere in six steps

Assumptions: plane-parallel and static; LTE ($S = B$); gray ($\kappa_\nu = \kappa$, in practice the Rosseland mean); radiative equilibrium with $F = \sigma T_{\rm eff}^4$.

1. **Radiative equilibrium:** $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$ becomes $\kappa(J - B) = 0$, so $\boxed{J = B = S}$. Mathematically this is a pure-scattering atmosphere (the Milne problem).
2. **Constant flux:** $dH/d\tau = J - B = 0$, so $H = \sigma T_{\rm eff}^4 / 4\pi$.
3. **Exact:** $dK/d\tau = H$, so $K = H\tau + \text{const}$.
4. **Eddington closure** $K = J/3$ gives $J = 3H(\tau + C)$. *(First approximation.)*
5. **Boundary condition** $J(0) = 2H$ gives $C = 2/3$. *(Second approximation.)*
6. **Temperature:** with $J = \sigma T^4/\pi$,

$$\boxed{T^4(\tau) = \tfrac{3}{4} T_{\rm eff}^4 \left(\tau + \tfrac{2}{3}\right)}$$

Consequence	Value
$T(\tau = 2/3)$	$= T_{\rm eff}$: the photosphere
$T(0)$	$2^{-1/4} T_{\rm eff} = 0.841\,T_{\rm eff}$: the atmosphere never cools to zero
$\tau \gg 1$	$T^4 \approx \tfrac{3}{4} T_{\rm eff}^4 \tau$: the diffusion regime
limb darkening	S is linear, so Eddington–Barbier is exact: $I(\mu)/I(1) = (2 + 3\mu)/5$

The exact solution (Milne 1921; Hopf 1930–34; Chandrasekhar 1950). Without approximations, Step 1 gives Milne's integral equation $J = \Lambda[J]$, whose solution is

$$\boxed{T^4 = \tfrac{3}{4} T_{\rm eff}^4 \left[\tau + q(\tau)\right]}$$

where the Hopf function q rises from $q(0) = 1/\sqrt{3} = 0.5774$ to $q(\infty) = 0.7104$; Eddington's 2/3 lies between. The exact surface temperature is $0.8114\,T_{\rm eff}$ (Eddington is 3.6 % too hot). The exact limb darkening is $I(\mu) \propto H(\mu)$, Chandrasekhar's H-function with $H(1) = 2.9078$.

μ	0	0.1	0.2	0.5	1
exact $I(\mu)/I(1)$	0.3439	0.4290	0.4988	0.6922	1
Eddington $(2+3\mu)/5$	0.400	0.460	0.520	0.700	1

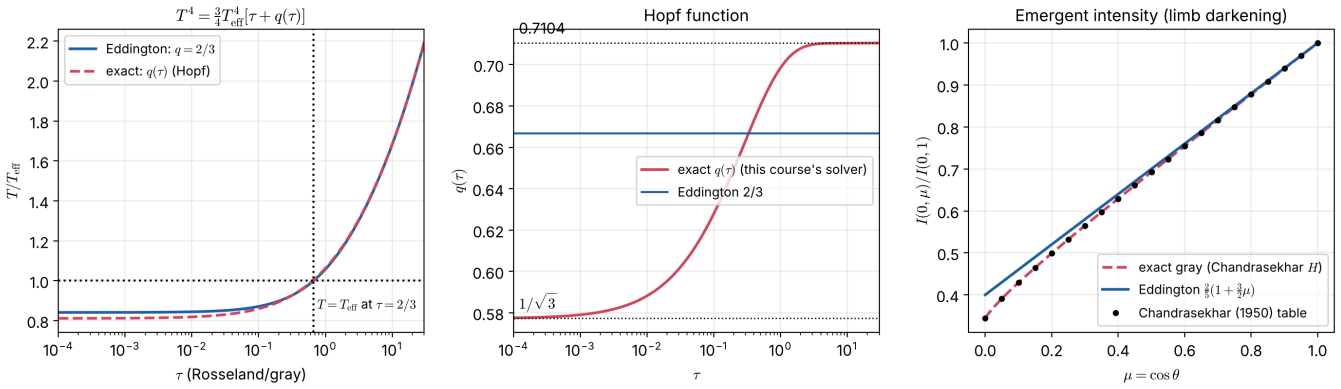


Figure 4.1 — Left: $T(\tau)$, Eddington versus exact; they differ only for $\tau \lesssim 1$. Centre: the Hopf function from our numerical solution of Milne's equation (`scripts/lte_solvers.py`). Right: limb darkening. The exact H-function law agrees with Chandrasekhar's table to 4 decimals; Eddington's is too bright at the limb.

Our solver builds the Λ matrix, attaches the known deep asymptote, and solves $(\mathbf{1} - \mathbf{\Lambda})\mathbf{S} = \mathbf{b}$. It reproduces $q(0) = 0.5773$, and the emergent flux integrated over all wavelengths equals $\sigma T_{\rm eff}^4$ to 0.007 %.

4.2 Gray opacity, non-gray spectrum

Each frequency samples $B_\nu(T(\tau))$ over a range of depths:

$$F_\nu(0) = 2\pi \int_0^\infty B_\nu\big(T(\tau)\big) E_2(\tau)\, d\tau.$$

Because B_ν depends very steeply on T in the Wien regime, hot deep layers dominate the UV. The result is a **UV excess** relative to $\pi B_\nu(T_{\rm eff})$, and limb darkening that is **stronger in the blue**.

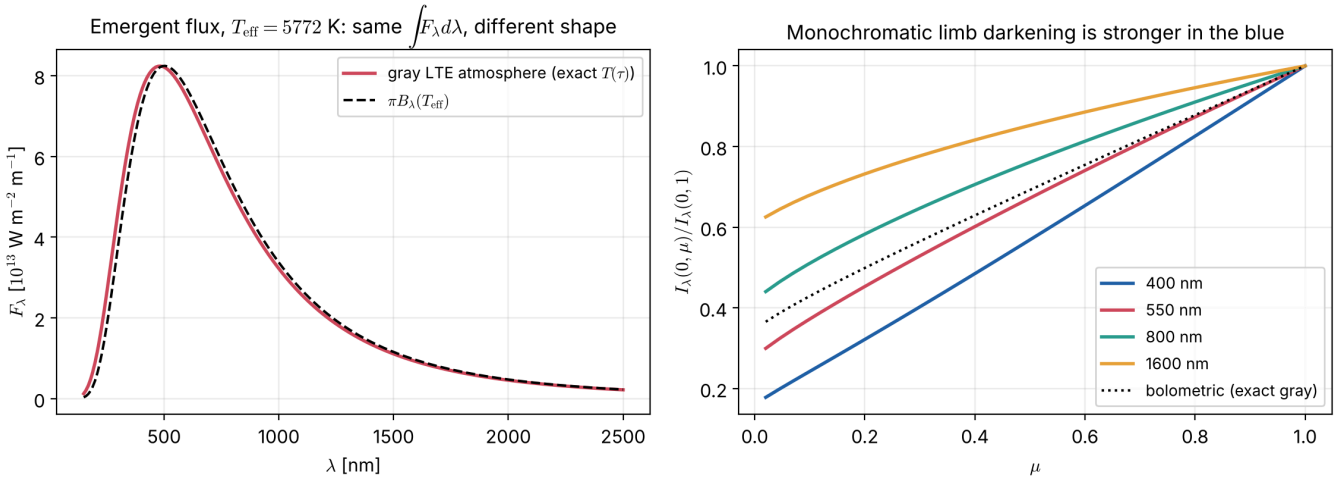


Figure 4.2 — Left: emergent flux of the exact gray atmosphere ($T_{\rm eff} = 5772$ K) compared with $\pi B_\lambda(T_{\rm eff})$. Right: limb darkening at 400, 550 and 800 nm, with $I(0.1)/I(1)$ = 0.242, 0.372 and 0.510.

Adding pressure. Hydrostatic equilibrium in optical depth, $dP/d\tau = g/\kappa$, gives $P = g\tau/\kappa$ for constant κ, and then $\rho = P\mu m_H/kT$. Together with $T(\tau)$ this is a complete model specified by $T_{\rm eff}$, g and composition, as Collins' five assumptions promised. Lecture SA5 does it with a realistic, pressure-dependent opacity, following Novotny.

4.3 How lines form in LTE

In LTE both continuum and line emit thermally, so $S_\nu = B_\nu(T)$ at every frequency. **What differs is the depth scale.** With $\eta_\nu = \kappa_l\phi/\kappa_c$, $d\tau_\nu = (1+\eta_\nu)\,d\tau_c$, and Eddington–Barbier gives $F_\nu \approx \pi B_\nu(T(\tau_\nu = 2/3))$:

- continuum: $\tau_\nu = 2/3$ at $\tau_c = 2/3$, deep and hot;
- line centre: $\tau_\nu = 2/3$ at $\tau_c = (2/3)/(1+\eta)$, high and cool.

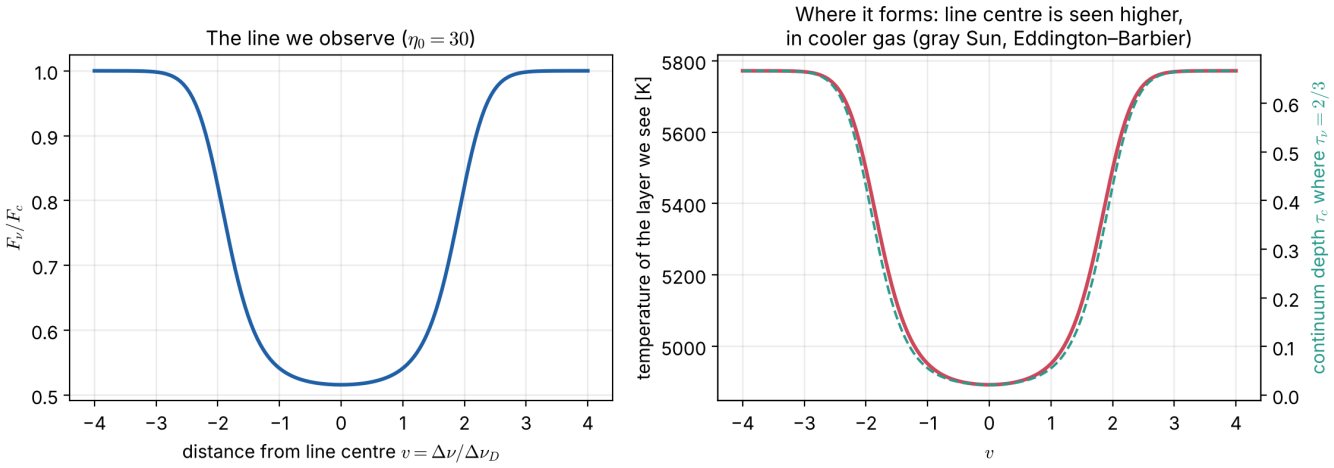


Figure 4.3 — Left: a line with $\eta_0 = 30$ in a gray solar atmosphere. Right: the temperature of the layer we see at each frequency (red) and the continuum optical depth where $\tau_\nu = 2/3$ (teal). The line centre is seen at a continuum optical depth 31 times smaller, in gas some 900 K cooler. ***An LTE absorption line is a map of the temperature gradient.***

This is the continuous version of MIT's two-layer star (Lecture RT3). An isothermal atmosphere gives **no lines**; a temperature rising outward gives **emission**, as in the Ca II H & K emission reversals of the chromosphere.

4.4 Milne–Eddington lines and the curve of growth

Milne–Eddington assumptions: LTE, $B = B_0(1+\beta\tau_c)$, and η_ν constant with depth. Then B is linear in τ_ν, Eddington–Barbier is exact, and

$$\boxed{r_\nu = \frac{F_\nu}{F_c} = \frac{1+\dfrac{2\beta}{3(1+\eta_\nu)}}{1+\dfrac{2\beta}{3}}} \qquad r \xrightarrow{\eta\to\infty} \frac{1}{1+2\beta/3}.$$

The deepest possible line is set by the temperature gradient β: $\beta = 0$ gives no line. For the bolometric gray atmosphere $\beta = 3/2$, so the maximum depth is 0.5; at visible wavelengths β is larger, so lines are deeper.

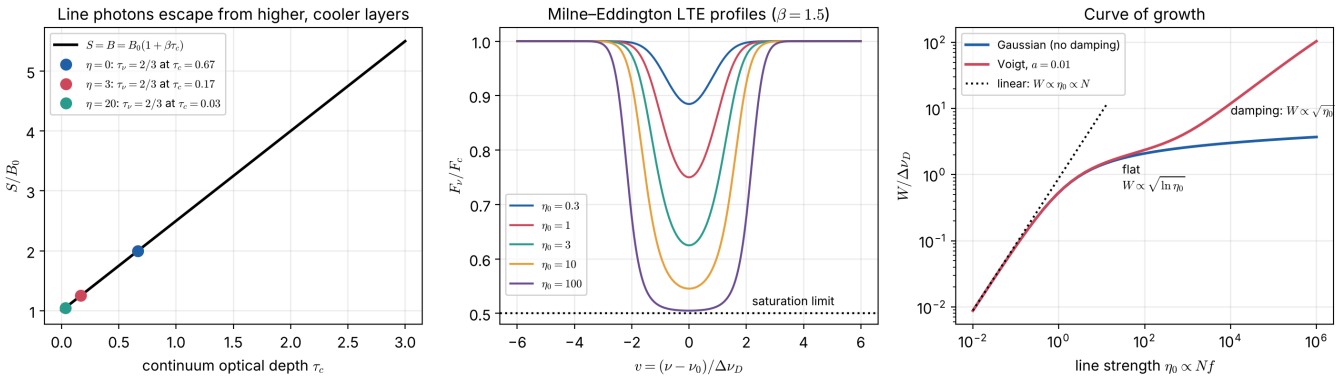


Figure 4.4 — Left: line photons escape from higher layers, where $S = B$ is smaller. Centre: Milne–Eddington profiles for growing η_0; the core saturates at a finite depth. Right: the curve of growth: linear, then flat, then square-root.

Equivalent width $W_\lambda = \int (1 - F_\lambda/F_c)\, d\lambda$. For a weak line,

$$\boxed{W_\lambda = \pi r_e \lambda^2 N_l f} \qquad \text{(independent of broadening).}$$

Part of the curve	Condition	W depends on	Use
linear	$\tau_0 \lesssim 1$	$W \propto Nf$	**abundances**
flat (saturated)	$1 \lesssim \tau_0 \lesssim 10^{3\text{–}4}$	$W \propto \Delta\nu_D \sqrt{\ln \tau_0}$	microturbulence ξ
damping	wings opaque	$W \propto \sqrt{Nf\gamma}$	gravity

Beyond LTE: for strong lines in thin upper layers, $S^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B$ falls below B, and cores are darker than LTE predicts (Lecture SA3).

Worked example 4.1 — The Sun as a gray atmosphere

$T_{\text{eff}} = 5772$ K. *Find T at several depths (Eddington) and the exact surface temperature.*

Solution.

τ	0	2/3	1	2	10
T (Eddington) [K]	4854	5772	6103	6864	9707

The exact surface temperature is $0.8114 \times 5772 = 4682$ K. The observed solar temperature minimum (~4400 K) is close; above it the chromosphere heats up, which radiative equilibrium cannot produce. The limb-darkening inversion of Lecture RT4 (5155 K at $\tau_{500} = 0$, 6300 K at $\tau_{500} = 1$) agrees to within the effects of non-grayness.

Worked example 4.2 — Milne–Eddington line depths

With $\beta = 1.5$, find the central residual flux for η_0 = 1, 10 and ∞.

Solution. $r = [1 + 1/(1+\eta)]/2$: r = 0.75, 0.545 and 0.5. Going from $\eta = 10$ to ∞ barely changes the core: the line is **saturated**, and W can now grow only in the wings.

Worked example 4.3 — An abundance from a weak line

A weak Fe I line at 500 nm has $W_\lambda = 22$ mÅ and $f = 0.1$. Find the absorbing column N_l.

Solution.

$$N_l = \frac{W_\lambda}{\pi r_e \lambda^2 f} = \frac{2.2 \times 10^{-12}}{8.853 \times 10^{-15} \times (5 \times 10^{-7})^2 \times 0.1} = 1.0 \times 10^{16}\ \mathrm{m}^{-2}.$$

The total iron abundance then follows from Saha–Boltzmann (the fraction of Fe in that level of Fe I) and the hydrogen column above $\tau \approx 2/3$. **LTE enters at the Saha–Boltzmann step.**

Common misconceptions

1. **"$T_{\rm eff}$ is the temperature of the stellar surface."** It is the temperature at $\tau \approx 2/3$; the top is about 19 % cooler.
2. **"Gray opacity gives a black-body spectrum."** Each frequency averages B_ν over a range of temperatures; the UV shows an excess.
3. **"Absorption lines need cold gas in front of the star."** They need only a temperature that falls outward within the atmosphere.
4. **"Saturated lines are useless for abundances."** They are less sensitive to abundance, but they constrain microturbulence (flat part) and gravity (damping part).

Lecture summary

- Gray + LTE + radiative equilibrium: $J = B$, H constant, K linear in τ. Eddington gives $T^4 = \frac{3}{4} T_{\rm eff}^4 (\tau + \frac{2}{3})$; the exact result replaces 2/3 by the Hopf function $q(\tau)$, from 0.577 to 0.710.
- $T(2/3) = T_{\rm eff}$, $T(0) = 0.811\, T_{\rm eff}$; limb darkening $(2 + 3\mu)/5$ (Eddington) or $H(\mu)$ (exact).
- Gray opacity still gives a non-Planckian spectrum, with a UV excess.
- LTE lines form because line opacity lifts $\tau_\nu = 2/3$ into cooler layers; no gradient, no lines.
- Milne–Eddington: $r = [1 + 2\beta/3(1+\eta)]/(1 + 2\beta/3)$. Curve of growth: linear (abundance), flat (ξ), damping (g).

Exam questions and answers — Lecture SA4

Q1. State the assumptions of the gray LTE atmosphere and derive $T^4 = \frac{3}{4} T_{\rm eff}^4 (\tau + \frac{2}{3})$, marking where approximations enter.

> **A1.** Plane-parallel, static, LTE, gray, radiative equilibrium with $F = \sigma T_{\rm eff}^4$. (i) RE with gray opacity: $J = B$. (ii) $dH/d\tau = J - B = 0$, so $H = F/4\pi$. (iii) $dK/d\tau = H$, so $K = H\tau + c$ (exact). **(iv) Eddington closure** $K = J/3$: $J = 3H(\tau + C)$. **(v) Boundary condition** $J(0) = 2H$: $C = 2/3$. (vi) $J = \sigma T^4/\pi$ and $H = \sigma T_{\rm eff}^4/4\pi$ give the result.

Q2. What is the Hopf function? Give $q(0)$ and $q(\infty)$, and the exact and Eddington surface temperatures.

A2. The exact solution of Milne's equation is $J = 3H[\tau + q(\tau)]$, where q rises from $1/\sqrt{3} = 0.577$ (the Hopf–Bronstein relation $J(0) = \sqrt{3}H$) to 0.7104. Then $T(0) = (\sqrt{3}/4)^{1/4} T_{\text{eff}} = 0.811\, T_{\text{eff}}$ exactly, against $2^{-1/4} = 0.841$ in the Eddington approximation, which replaces q by the constant 2/3.

Q3. Explain, using Eddington–Barbier, why the solar spectrum shows absorption lines. What would an isothermal LTE atmosphere show, and what does a chromosphere do?

A3. $F_\nu \approx \pi S_\nu(\tau_\nu = 2/3)$ with $S_\nu = B_\nu(T)$. Line opacity makes $\tau_\nu = 2/3$ occur higher up, where T and B_ν are lower, so the flux is lower: an absorption line. An isothermal atmosphere has the same B_ν at all depths, so $F_\nu = F_c$ and there are no lines. In a chromosphere T rises outward, so the cores of strong lines formed there appear in emission (e.g. the Ca II K_2 peaks).

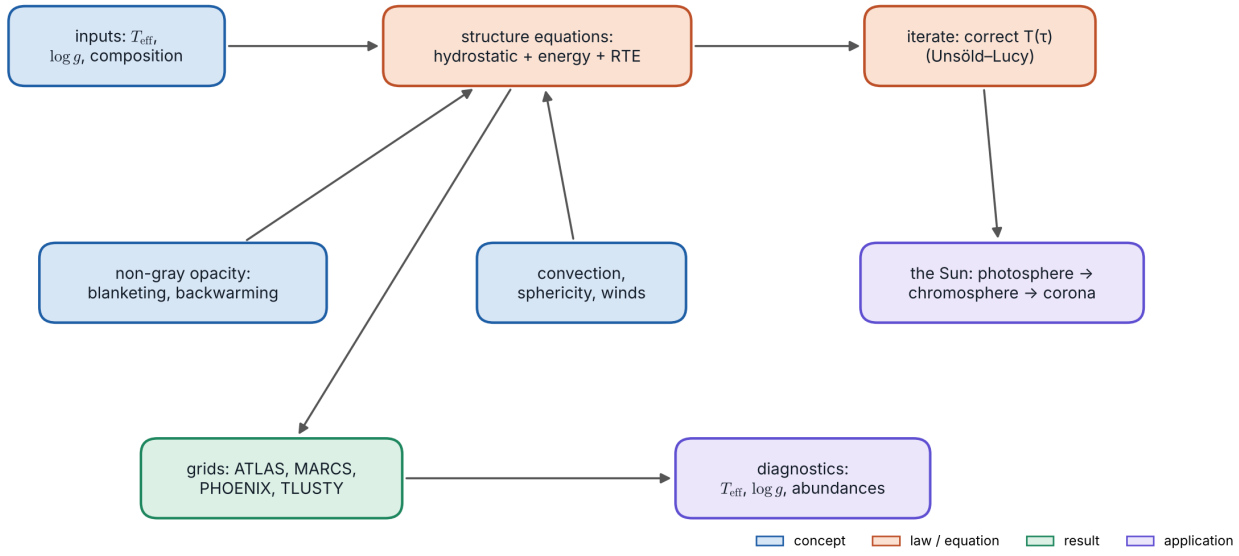
Q4. Sketch the curve of growth and explain each of its three parts. Which is best for abundances, and why?

A4. Linear: optically thin; each atom removes photons independently, so $W \propto Nf$. Flat: the core is saturated, and W grows only as the region with $\tau_\nu > 1$ widens through the Doppler core, $W \propto \Delta\nu_D \sqrt{\ln \tau_0}$. Damping: the Lorentzian wings become opaque, $W \propto \sqrt{Nf\gamma}$. The linear part is best for abundances because it does not depend on the poorly known ξ and γ.

Lecture SA5 — Real Model Atmospheres and the Sun

> **Sources:** E. Novotny, *Introduction to Stellar Atmospheres and Interiors* (1973), Ch. 4 (the equations for model atmospheres; convection; comparison of models) and Ch. 9 (a model solar atmosphere computed step by step); Collins §9.1d; Mihalas (1978) Ch. 7; Gray, *The Observation and Analysis of Stellar Photospheres*; Vernazza, Avrett & Loeser (1981); Asplund et al. (2009); Hubeny & Mihalas (2014).

SA5 concept map — real model atmospheres



*Concept map — three inputs ($T_{\rm eff}$, $\log g$, composition) feed the **structure equations**: hydrostatic equilibrium, energy conservation and radiative transfer. Real opacity is non-gray (blanketing, backwarming), and convection, sphericity and winds enter as corrections. The equations are iterated until consistent. The output is a grid of models, used to diagnose real stars, starting with the Sun.*

Learning goals

1. List the equations of a classical model atmosphere and explain why three parameters fix it.
2. Build a simple solar model by hand, following Novotny: $T(\tau)$, then $P(\tau)$, then a convection test.
3. Explain line blanketing, backwarming and surface cooling.
4. Describe the layers of the solar atmosphere and which need physics beyond radiative-equilibrium LTE.
5. Know when sphericity, winds and 3D effects matter, and name the main codes and diagnostics.

The big ideas in plain words

A model atmosphere is a self-consistent recipe. Give it $T_{\rm eff}$, $\log g$ and composition. It returns, depth by depth, T, P, ρ, ionisation, opacity and the radiation field, all agreeing with each other: the energy flows out at a constant rate, pressure balances gravity, and populations follow from T and P.

You can build one with a pencil. Novotny's recipe takes three steps: get $T(\tau)$ from the gray solution; integrate hydrostatic equilibrium for $P(\tau)$ using the opacity; and test whether the layers are stable against convection. Every modern code does the same thing, just with millions of frequencies and many iterations.

Lines push energy around. Thousands of lines block light in some windows. The energy squeezes through other windows and heats the deeper layers (*backwarming*), while lines cool the top layers.

The Sun's atmosphere gets hotter outward above a minimum. The temperature falls through the photosphere to about 4200 K, then rises in the chromosphere and jumps to millions of degrees in the corona. Radiative equilibrium cannot do that; waves and magnetic fields deposit energy there.

5.1 The structure equations

Equation	Form	Role
hydrostatic equilibrium	$dP_{\rm gas}/d\tau = g/\kappa - dP_{\rm rad}/d\tau$	pressure
energy conservation	$F_{\rm rad} + F_{\rm conv} = \sigma T_{\rm eff}^4$	temperature
radiative transfer	$\mu\, dI_\nu/d\tau_\nu = I_\nu - S_\nu$ at 10^4–10^6 frequencies	radiation field
convection	mixing length, where $\nabla_{\rm rad} > \nabla_{\rm ad}$	convective flux
equation of state	ideal gas + Saha–Boltzmann (LTE) or rate equations (NLTE)	populations, n_e, ρ
opacity	$\kappa_\nu(T, P, \text{composition})$	couples everything

In plane-parallel geometry g is constant and the flux is fixed by $T_{\rm eff}$. The radius appears nowhere, so $T_{\rm eff}$, $\log g$ **and composition determine the model.**

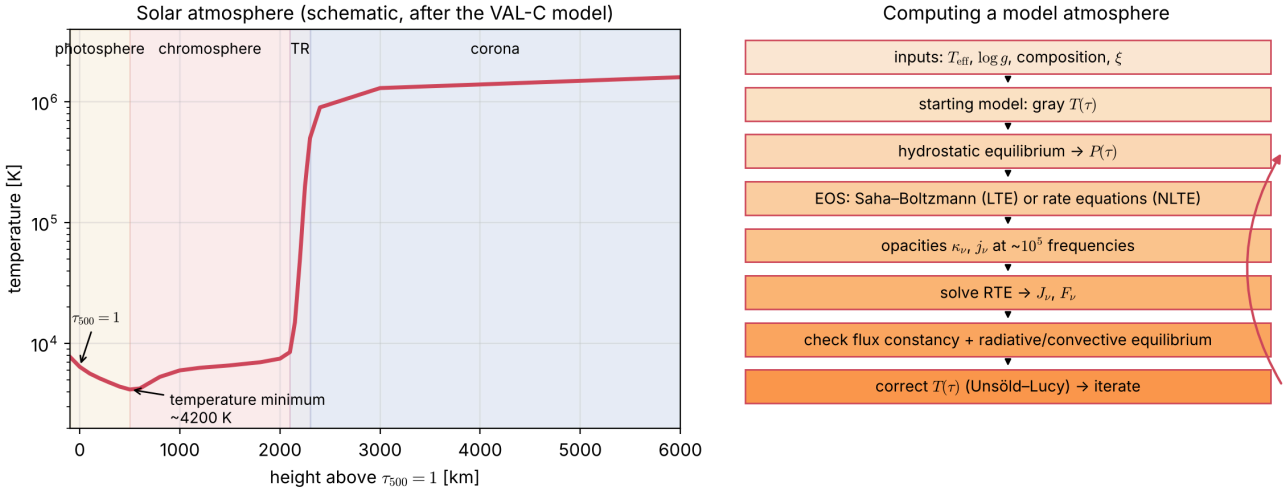


Figure 5.1 — Left: the layers of the solar atmosphere (schematic, after VAL-C): T falls to a minimum near 4200 K, rises through the chromosphere, jumps through the transition region and reaches 1–2 MK in the corona. Right: the iteration. Start from the gray $T(\tau)$ of Lecture SA4, compute the opacities and radiation field, test flux constancy, correct T (Unsöld–Lucy, Avrett–Krook, or complete linearization) and repeat.

5.2 A solar model atmosphere by hand (after Novotny, Ch. 9)

Novotny's problem: find T, P and ρ for $0.05 < \tau < 2$ in the Sun, given $T_{\rm eff} = 5800$ K and $g = 2.74 \times 10^4$ cm s^{-2} (cgs units, as in the book).

Step 1 — Temperature. Take the gray Eddington law (Lecture SA4):

$$T^4 = \tfrac{3}{4} T_{\rm eff}^4 \left(\tau + \tfrac{2}{3}\right), \qquad T = 4877\,(1 + 1.5\tau)^{1/4}\ \text{K}.$$

Step 2 — Pressure. Hydrostatic equilibrium in optical depth is $dP/d\tau = g/\kappa(P,T)$. Novotny reads the Rosseland mean κ from Vitense's (1951) tables. For $\theta = 5040/T$ between 0.9 and 1.0 and $\log P$ between 4 and 5, the tables are well fitted by

$$\kappa \approx 10^{-4.40} P^{0.8}\ \text{cm}^2\,\text{g}^{-1}.$$

With this power law the equation separates: $P^{0.8}\,dP = 10^{4.40} g\,d\tau$, so

$$\boxed{P^{1.8} = 1.8 \times 10^{4.40}\, g\,\tau} \qquad (P = 0 \text{ at } \tau = 0).$$

This analytic law provides the *starting values*. Novotny then integrates numerically with the full tabulated $\kappa(P,T)$, using finite differences (Table 9-2), to $\tau = 2$.

Step 3 — Convection test. The layer is stable if the actual gradient is shallower than the adiabatic one, $\nabla = d\ln T/d\ln P < \nabla_{\rm ad} = 0.4$, i.e.

$$\frac{d\ln P}{d\ln T} > 2.5 \quad \text{(stable)}.$$

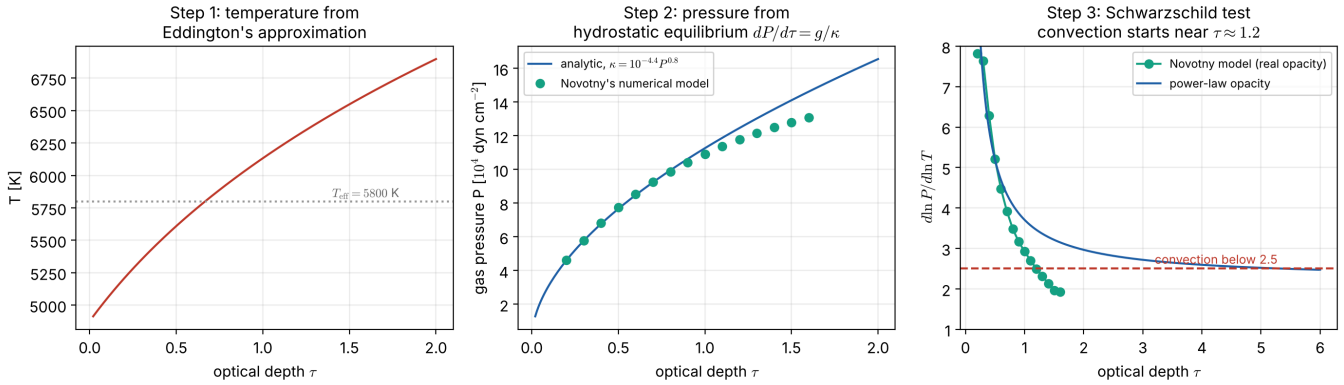


Figure 5.2 — Building the Sun's atmosphere in three steps (`scripts/sa_physics.py`). Left: $T(\tau)$ from Eddington's approximation. Centre: $P(\tau)$; the analytic power-law solution (line) agrees with Novotny's numerical integration with real opacities (dots) to 3 % at $\tau = 1$. Right: the Schwarzschild test. With real opacity, $d\ln P/d\ln T$ drops below 2.5 near $\tau \approx 1.2$, where convection begins. With a power-law opacity that ignores the temperature dependence, convection would start only near $\tau \approx 5$.

The lesson of Step 3. In the gray, constant-κ model, $\nabla = \tau/[4(\tau + 2/3)] < 0.25$, so convection never starts. In the real Sun the H$^-$ opacity rises steeply with T; κ grows inward, the pressure gradient g/κ shrinks, $d\ln P/d\ln T$ falls, and convection sets in just below the photosphere. Hydrogen ionisation lowers $\nabla_{\rm ad}$ further. The top of the convection zone is what we see as **granulation**: cells about 1000 km across that last 5–10 minutes. 1D models use mixing-length theory plus a microturbulence parameter; **3D radiation-hydrodynamics** (Stagger, CO5BOLD, MURaM, Bifrost) simulates convection directly.

5.3 Real opacity: blanketing across the spectral sequence

Type	$T_{\rm eff}$ [K]	Dominant continuum opacity	Spectral features
O, early B	30 000–50 000	electron scattering; He bound–free	He II lines; winds; NLTE essential
late B, A	10 000–20 000	H I bound–free	strongest Balmer lines and jump near A0
F, G, K	4000–7500	$\mathbf{H^-}$	metal lines, G band, Ca II H & K
M	2500–4000	H^- + molecules	TiO, VO, H_2O bands
L, T	< 2500	molecules, dust clouds	CH_4 and H_2O bands

Line blanketing has three effects: **blocking** (less UV flux, redder colours); **backwarming** (the same total flux must get through, so the deeper layers heat up); and **surface cooling** (lines radiate efficiently from the top, lowering the boundary temperature below the gray value).

5.4 The solar atmosphere, layer by layer

Layer	Height [km]	T	Physics	Diagnostics
photosphere	0–500	6400 → 4200 K	radiative equilibrium, mostly LTE, convective overshoot	visible continuum, weak lines, limb darkening
temperature minimum	~500	~4200 K		UV near 160 nm, far IR
chromosphere	500–2100	6000–20 000 K	wave and magnetic heating; NLTE	Hα, Ca II H & K, Mg II h & k, mm continuum
transition region	~2100–2300	$10^4 \rightarrow 10^6$ K	conduction-dominated	C IV, Si IV, O V
corona	> 2300	1–2 MK	magnetic heating; coronal approximation	EUV and X-ray lines (Fe IX–XVI); white-light K-corona

Only the photosphere is close to radiative-equilibrium LTE. The rise above the temperature minimum needs non-radiative heating, and its lines need NLTE (Lecture SA3). In LTE such a rise would give emission cores in strong lines: the thermal-inversion effect of MIT's toy model (Lecture RT3).

5.5 Beyond plane-parallel and static; codes; diagnostics

- **Sphericity** matters when the atmosphere's thickness, about $10H$ with $H = kT/(\mu m_H g)$, exceeds a few per cent of R, as in giants and supergiants.
- **Winds**: hot stars drive winds by radiation pressure on lines (Castor, Abbott & Klein 1975), producing P Cygni profiles. These are modelled with the Sobolev approximation or in the comoving frame.
- **3D and NLTE** analyses lowered the solar oxygen abundance by about 0.2 dex (Asplund et al. 2009).

Code or grid	Type	Use
ATLAS9/12 (Kurucz)	1D LTE, line-blanketed	A–K stars
MARCS	1D LTE, plane-parallel or spherical, molecules	F–M stars, giants
PHOENIX	1D/3D spherical, NLTE capable	cool stars, brown dwarfs, novae, supernovae
TLUSTY	1D NLTE	O and B stars, white dwarfs, disks
CMFGEN, PoWR, FASTWIND	spherical NLTE with winds	massive stars, Wolf–Rayet stars
Stagger, CO5BOLD, MURaM, Bifrost	3D radiation-(M)HD	granulation, the solar atmosphere
SYNTHE, MOOG, Turbospectrum, SME	spectrum synthesis	abundances

Parameter	Main diagnostics
$T_{\rm eff}$	spectral energy distribution, infrared flux method, Balmer wings (F–G), Fe I excitation balance, interferometry
$\log g$	pressure-broadened wings (Lecture SA2), Fe I/Fe II ionisation balance, asteroseismology, binaries
abundances	curve of growth and spectrum synthesis (Lecture SA4), with NLTE and 3D corrections
microturbulence	flat part of the curve of growth
magnetic field	Zeeman splitting and Stokes V (Lecture RT5)

Worked example 5.1 — Numbers from the hand-built Sun

Using Novotny's recipe, find T and P at $\tau = 0.2$ and $\tau = 1$, the density at $\tau = 1$ (mean molecular weight 1.3), and the depth where convection begins.

Solution.

- $\tau = 0.2$: $T = 4877(1.3)^{1/4} = 5208$ K; $P = (1.8 \times 10^{4.40} \times 2.74 \times 10^4 \times 0.2)^{1/1.8} = 4.61 \times 10^4$ dyn cm^{-2}, exactly Novotny's starting value (4.607).
- $\tau = 1$: $T = 6133$ K. The analytic law gives 1.13×10^5 dyn cm^{-2}; Novotny's integration gives 1.091×10^5 (= 1.09×10^4 Pa).
- Density: $\rho = P\mu m_H / kT = 1.09 \times 10^4 \times 1.3 \times 1.674 \times 10^{-27} / (1.381 \times 10^{-23} \times 6133) = 2.8 \times 10^{-4}$ kg m^{-3}, close to detailed solar models.
- Convection: from Novotny's table, $d\ln P / d\ln T$ = 2.70 at $\tau = 1.1$ and 2.49 at $\tau = 1.2$ (his 2.68 and 2.48), so convection begins just above $\tau = 1.2$. Novotny adds that radiation still carries most of the flux there.

Worked example 5.2 — When is an atmosphere plane-parallel?

Find $H = kT/(\mu m_H g)$ ($\mu = 1.3$) for the Sun, a red supergiant ($T = 4000$ K, $g = 0.01$ m s^{-2}, $R = 500 R_\odot$) and a white dwarf ($T = 10^4$ K, $g = 10^6$ m s^{-2}).

Solution.

- Sun: $H = 134$ km, so $10H/R_\odot = 0.2\,\%$: plane-parallel is excellent.
- Supergiant: $H = 2.5 \times 10^9$ m $= 3.6\,R_\odot$, so $10H/R = 7\,\%$: spherical models are needed.
- White dwarf: $H \approx 64$ m. The whole atmosphere is a few hundred metres thick, with enormous pressure broadening.

Worked example 5.3 — Energy budget of the chromosphere

The chromosphere radiates ~4×10^3 W m^{-2} and the quiet corona ~3×10^2 W m^{-2}. Compare with the photospheric flux.

Solution. $\sigma T_{\rm eff}^4 = 6.3 \times 10^7$ W m^{-2}, so the ratios are 6×10^{-5} and 5×10^{-6}. Only about 10^{-4} of the energy needs to be deposited non-radiatively. The *temperature* is high because the density is so low that very few particles must radiate this power.

Common misconceptions

1. **"The photosphere is the hottest visible layer."** The chromosphere and corona are hotter but faint in visible light, because their density is low.
2. **"Line blanketing only removes light."** The blocked energy is redistributed: it emerges elsewhere and heats deeper layers.
3. **"Model atmospheres need the stellar radius."** Plane-parallel models need only $T_{\rm eff}$, $\log g$ and composition.
4. **"1D LTE models are always good enough for abundances."** 3D and NLTE effects change many abundances by 0.1–0.3 dex, especially in metal-poor stars and giants.

Lecture summary

- A model atmosphere solves hydrostatic equilibrium, energy conservation, radiative transfer, the equation of state and the opacities together; $T_{\rm eff}$, $\log g$ and composition fix it.
- Novotny's hand-built Sun: gray $T(\tau)$, then $P^{1.8} = 1.8 \times 10^{4.4} g\tau$ as starting values, then numerical integration and the Schwarzschild test. Convection begins near $\tau \approx 1.2$ because H$^-$ opacity rises steeply with T.
- Non-gray opacity causes blocking, backwarming and surface cooling; the dominant opacity runs from electron scattering (O) to H I (A) to H$^-$ (F–K) to molecules (M).
- The Sun: photosphere → temperature minimum (~4200 K) → chromosphere → transition region → corona (1–2 MK), with non-radiative heating and NLTE above the minimum.

Exam questions and answers — Lecture SA5

Q1. List the equations of a classical 1D LTE model atmosphere, and explain why $T_{\rm eff}$, $\log g$ and composition determine it.

> **A1.** Hydrostatic equilibrium $dP/d\tau = g/\kappa$ (plus radiation and turbulent pressure); energy conservation $F_{\rm rad} + F_{\rm conv} = \sigma T_{\rm eff}^4$; radiative transfer at many frequencies; mixing-length convection; the equation of state

with Saha–Boltzmann populations; and opacities $\kappa_\nu(T, P, \text{composition})$. In plane-parallel geometry g is constant, the flux is fixed by T_{eff}, and the radius appears nowhere.

Q2. Following Novotny, show that if $\kappa = \kappa_0 P^a$, hydrostatic equilibrium gives $P^{1+a} = (1+a) g\tau/\kappa_0$. Evaluate P at $\tau = 0.2$ for the Sun with $\kappa_0 = 10^{-4.4}$ and $a = 0.8$ (cgs).

A2. $dP/d\tau = g/(\kappa_0 P^a)$, so $P^a\, dP = (g/\kappa_0)\, d\tau$. Integrating with $P(0) = 0$ gives $P^{1+a}/(1+a) = g\tau/\kappa_0$. Numerically, $P^{1.8} = 1.8 \times 10^{4.4} \times 2.74 \times 10^4 \times 0.2 = 2.48 \times 10^8$, so $P = 4.6 \times 10^4$ dyn cm^{-2} (4.6 kPa).

Q3. Why is the gray, constant-opacity atmosphere stable against convection, while the real Sun becomes convective just below its photosphere?

A3. With constant κ, $P \propto \tau$ and $T^4 \propto \tau + 2/3$, so $\nabla = d\ln T/d\ln P = \tau/[4(\tau + 2/3)] < 0.25 < \nabla_{\text{ad}} = 0.4$: stable. In the Sun the H$^-$ opacity rises steeply with T, so κ grows inward and $dP/d\tau = g/\kappa$ shrinks. $d\ln P/d\ln T$ then drops below 2.5 (i.e. $\nabla > 0.4$) near $\tau \approx 1.2$ (Novotny's model). Hydrogen ionisation also lowers ∇_{ad}.

Q4. Describe the temperature structure of the solar atmosphere from photosphere to corona, and say which parts radiative-equilibrium LTE can describe.

A4. T falls from about 6400 K at $\tau_{500} = 1$ to a minimum of about 4200 K some 500 km up (photosphere: radiative equilibrium, mostly LTE). It then rises through the chromosphere (6000–20 000 K), jumps in the transition region near 2100–2300 km, and reaches 1–2 MK in the corona. Only the photosphere is close to radiative-equilibrium LTE; the chromosphere and corona need non-radiative heating, and their lines need NLTE.

Final Examination — Course B: Stellar Atmospheres (with complete solutions)

Format suggestion: 3 hours. Part A (20 %): short answers. Part B (50 %): answer any 6 of the 8 problems. Part C (15 %): one essay. Part D (15 %): classification. Solutions follow each question; numbers were checked with `scripts/lte_solvers.py` and `scripts/sa_physics.py`. Constants: $h = 6.626 \times 10^{-34}$ J s, $k = 1.381 \times 10^{-23}$ J K^{-1}, $c = 2.998 \times 10^{8}$ m s^{-1}, $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}, $m_e = 9.109 \times 10^{-31}$ kg, $m_H = 1.674 \times 10^{-27}$ kg, $r_e = 2.818 \times 10^{-15}$ m, $\sigma_T = 6.652 \times 10^{-29}$ m^2, $R_\odot = 6.957 \times 10^{8}$ m, 1 pc $= 3.086 \times 10^{16}$ m, 1 eV $= 1.602 \times 10^{-19}$ J. $\int_0^\infty E_n(t)\,dt = \dfrac{1}{n}$, $\int_0^\infty t\,E_n(t)\,dt = \dfrac{1}{n+1}$, $E_n(0) = \dfrac{1}{n-1}$.

Part A — Short answers (2 marks each)

A1. True or false: “In LTE the specific intensity equals the Planck function.”

False. In LTE the **source function** equals $B_\nu(T)$. $I_\nu = B_\nu$ only in full TE, or deep inside an optically thick, isothermal region.

A2. What is the exact boundary temperature of a gray atmosphere in units of $T_{\rm eff}$?

$(\sqrt{3}/4)^{1/4} = 0.811$.

A3. Why do Balmer lines peak in A stars?

Boltzmann excitation into $n = 2$ increases with T, while Saha ionisation removes neutral H above about 10^4 K. The product peaks near 9900 K.

A4. Name the integral equation of the gray LTE atmosphere.

Milne’s equation, $J(\tau) = \Lambda_\tau[J] = \tfrac{1}{2}\int_0^\infty J(t) E_1|t-\tau|dt$.

A5. Why are NLTE effects strongest in strong resonance lines?

They form in high, low-density layers where $C \ll A$ ($\varepsilon \ll 1$), and the thermalisation depth $1/\sqrt{\varepsilon}$ is very

large.

A6. What does a positive β in $B = B_0(1 + \beta\tau)$ imply for limb darkening and for line formation in LTE?

Temperature increasing inward: limb darkening, and absorption lines.

A7. What is the threshold wavelength of H^- bound–free absorption, and why?

1.64 µm: the extra electron is bound by only 0.754 eV, and $\lambda = hc/0.754\,\mathrm{eV}$.

A8. Write the Klein–Rosseland relation between collisional excitation and de-excitation.

$q_{lu}/q_{ul} = (g_u/g_l)\, e^{-\Delta E/kT}$.

A9. In the coronal approximation, how does the upper-level population scale with electron density?

$n_u/n_l = n_e q_{lu}/A_{ul} \propto n_e$ (far below the Boltzmann value).

A10. Which part of a Voigt profile measures pressure, and what is its asymptotic form?

The far wings: $H(a, v) \approx a/(\sqrt{\pi} v^2)$, with $a \propto \gamma$, which grows with the perturber density.

Part B — Problems

B1. Kirchhoff's law and LTE (8 marks)

a. Using the Einstein coefficients, write j_ν and α_ν for a line. (b) Show that if the level populations obey Boltzmann's law at temperature T, then $S_\nu = B_\nu(T)$. (c) Explain why this result does not require $I_\nu = B_\nu$.

Solution. (a) $j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi$, $\alpha_\nu = \frac{h\nu}{4\pi}\phi(n_1 B_{12} - n_2 B_{21})$. (b) $S = \frac{n_2 A_{21}}{n_1 B_{12} - n_2 B_{21}} = \frac{A_{21}/B_{21}}{\frac{n_1 g_2}{n_2 g_1} - 1}$ (using $g_1 B_{12} = g_2 B_{21}$). Boltzmann gives $\frac{n_1 g_2}{n_2 g_1} = e^{h\nu/kT}$, and $A_{21}/B_{21} = 2h\nu^3/c^2$, so $S = B_\nu(T)$. (c) Only the **populations** (a property of the matter, set by collisions) were used. The radiation field does not appear, so $S = B$ holds whenever the matter is in LTE, whatever I_ν is.

B2. Saha equation (8 marks)

Pure hydrogen has $n_e = 10^{20}$ m^{-3}. Find the ionised fraction at (a) $T = 6000$ K and (b) $T = 8000$ K. ($U_\mathrm{I} = 2$, $U_\mathrm{II} = 1$, $\chi = 13.6$ eV.)

Solution. $\frac{N_{\rm II}}{N_{\rm I}} = \frac{1}{n_e}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi/kT}$ (the factor $2U_{\rm II}/U_{\rm I} = 1$). (a) $T = 6000$ K: $(\dots)^{3/2} = 2.415\times10^{21}\times 6000^{1.5} = 1.12\times10^{27}$ m^{-3}; $\chi/kT = 26.3$, so $e^{-26.3} = 3.8\times10^{-12}$. Ratio $= 1.12\times10^{27}\times3.8\times10^{-12}/10^{20} = 4.2\times10^{-5}$: essentially neutral. (b) $T = 8000$ K: $(\dots)^{3/2} = 1.73\times10^{27}$; $\chi/kT = 19.7$, so $e^{-19.7} = 2.7\times10^{-9}$. Ratio $= 0.047$; ionised fraction $= 0.045$. A 33 % rise in T increases ionisation by a factor of about 1000: Saha ionisation is extremely sensitive to temperature.

B3. The gray LTE atmosphere — full derivation (12 marks)

a. Starting from the plane-parallel RTE with $S = B$ and gray opacity, show that radiative equilibrium implies $J = B$ and H = constant. (b) Using the Eddington approximation, derive $T(\tau)$. (c) For $T_{\rm eff} = 10\,000$ K, tabulate T at $\tau = 0, 2/3, 1, 4$. (d) Derive the emergent intensity $I(0,\mu)$ and the limb-darkening law. (e) The exact surface temperature is $0.811\,T_{\rm eff}$. What is the percentage error of the Eddington value?

Solution. (a) RE: $\int \kappa_\nu (J_\nu - B_\nu)d\nu = 0$, which with $\kappa_\nu = \kappa$ gives $J = B$. The zeroth moment $dH/d\tau = J - S = J - B = 0$ gives H = const $= \sigma T_{\rm eff}^4/4\pi$. (b) $dK/d\tau = H \Rightarrow K = H\tau +$ const; $K = J/3 \Rightarrow J = 3H(\tau + C)$; $J(0) = 2H \Rightarrow C = 2/3$. $J = B = \sigma T^4/\pi$ gives $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$. (c) $T(0) = 8409$ K, $T(2/3) = 10\,000$ K, $T(1) = 10\,574$ K, $T(4) = 13\,678$ K. (d) $S = 3H(\tau + 2/3)$ is linear, so $I(0,\mu) = 3H(\mu + 2/3) = \frac{3\sigma T_{\rm eff}^4}{4\pi}(\mu + \frac{2}{3})$, and $I(\mu)/I(1) = (2 + 3\mu)/5$. (e) $(0.841 - 0.811)/0.811 = 3.7\,\%$.

B4. Milne–Eddington line in LTE (10 marks)

In an LTE atmosphere $B_\nu = B_0(1 + \beta\tau_c)$, and a line has a constant ratio $\eta_\nu = \eta_0 e^{-v^2}$ of line to continuum opacity. (a) Derive F_ν/F_c. (b) For $\beta = 3$, find the central depth for $\eta_0 = 0.1, 1, \infty$. (c) Show that for weak lines the equivalent width is proportional to η_0, and find the constant (in units of $\Delta\nu_D$).

Solution. (a) $\tau_\nu = (1 + \eta_\nu)\tau_c$, so B is linear in τ_ν and E–B is exact: $r = \frac{1 + 2\beta/[3(1+\eta)]}{1 + 2\beta/3}$. (b) $\beta = 3$: $1 + 2\beta/3 = 3$. $\eta_0 = 0.1$: $r = (1 + 2/1.1)/3 = 0.939$, depth 0.061. $\eta_0 = 1$: $r = 2/3$, depth 0.333. $\eta_0 \to \infty$: $r = 1/3$, depth 0.667. (c) For $\eta \ll 1$: $1 - r \approx \frac{2\beta/3}{1+2\beta/3}\eta_\nu$, so $W/\Delta\nu_D = \int (1 - r)dv = \frac{2\beta/3}{1+2\beta/3}\eta_0\sqrt{\pi}$. For $\beta = 3$: $W/\Delta\nu_D = \frac{2}{3}\sqrt{\pi}\,\eta_0 = 1.18\,\eta_0$.

B5. Hydrogen bound–free opacity and the Balmer jump (8 marks)

a. Compute Kramers' cross-section ($g_{\rm bf} = 1$) at the Balmer edge. (b) Give the wavelengths of the Paschen and Brackett edges. (c) Explain why the Balmer jump is large in A stars and small in the Sun.

Solution. (a) $\nu_2 = 3.288\times10^{15}/4 = 8.22\times10^{14}$ Hz; $\sigma = 2.815\times10^{29}/(2^5\nu_2^3)$ cm^2 $= 1.6\times10^{-17}$ cm^2 $= 1.6\times10^{-21}$ m^2. (b) $\lambda_n = 91.13\,n^2$ nm: Paschen 820 nm, Brackett 1458 nm. (c) Near 9000–10 000 K, hydrogen bound–free from $n = 2$ dominates the continuum; blueward of 364.7 nm the opacity jumps

by a factor of about 26, we see higher and cooler layers, and the flux drops. In the Sun $\mathrm{H^-}$ dominates by a factor of about 100 and has no edge at 364.7 nm, so the hydrogen jump is diluted.

B6. The negative hydrogen ion (8 marks)

Using Saha's equation with $\chi = 0.754$ eV, $U(\mathrm{H^-}) = 1$, $U(\mathrm{H\,I}) = 2$, compute $n(\mathrm{H^-})/n(\mathrm{H\,I})$ for $n_e = 10^{19}$ $\mathrm{m^{-3}}$ at (a) 6000 K and (b) 5000 K. Explain the trend and why metal-poor stars have less $\mathrm{H^-}$ opacity.

Solution. $n(\mathrm{H^-})/n(\mathrm{H\,I}) = n_e / \left[4(2\pi m_e kT/h^2)^{3/2} e^{-0.754\,\mathrm{eV}/kT}\right]$. (a) 6000 K: $(\dots)^{3/2} = 1.12 \times 10^{27}$ $\mathrm{m^{-3}}$, $e^{-1.458} = 0.233$, ratio $= 9.6 \times 10^{-9}$. (b) 5000 K: 8.5×10^{26}, $e^{-1.750} = 0.174$, ratio $= 1.7 \times 10^{-8}$. At fixed n_e, cooler gas holds more $\mathrm{H^-}$ (the weak binding is less easily broken). In a real atmosphere, however, n_e itself falls steeply with T. Since the electrons come from metals, metal-poor stars have lower n_e, hence less $\mathrm{H^-}$ and lower opacity.

B7. Transition rates and critical density (10 marks)

A forbidden line has $A_{ul} = 0.01$ $\mathrm{s^{-1}}$, $\Omega = 1$, $g_u = 3$, at $T = 8000$ K. (a) Compute q_{ul} and n_{crit}. (b) Find $(n_u/n_l)/(n_u/n_l)_{\mathrm{LTE}}$ at $n_e = 10^{10}$ $\mathrm{m^{-3}}$. (c) Show that in the coronal limit the ionisation balance is independent of density.

Solution. (a) $q_{ul} = 8.629 \times 10^{-12}\Omega/(g_u\sqrt{T}) = 8.629 \times 10^{-12}/(3 \times 89.4) = 3.2 \times 10^{-14}$ $\mathrm{m^3\ s^{-1}}$; $n_{\mathrm{crit}} = A/q_{ul} = 3.1 \times 10^{11}$ $\mathrm{m^{-3}}$. (b) With $x = n_e/n_{\mathrm{crit}} = 0.032$, the ratio is $x/(1+x) = 0.031$: the upper level holds 3 % of its LTE population. (c) Coronal limit: $n_i n_e q_{\mathrm{ion}}(T) = n_{i+1} n_e \alpha_{\mathrm{rec}}(T)$; n_e cancels, so $n_{i+1}/n_i = q_{\mathrm{ion}}/\alpha_{\mathrm{rec}}$ depends only on T.

B8. Broadening of the Na D line (8 marks)

For Na I D at 589 nm in the solar photosphere, take $T = 5772$ K, $m = 23$ u, $\xi = 1$ km $\mathrm{s^{-1}}$ and total damping $\gamma = 1.5 \times 10^9$ $\mathrm{s^{-1}}$. Compute (a) the Doppler width in mÅ, (b) the damping parameter a, (c) the distance from line centre (in mÅ) where the Lorentzian wing equals the Gaussian core.

Solution. (a) $v_D = \sqrt{2kT/m + \xi^2} = \sqrt{4.17 \times 10^6 + 10^6} = 2.27$ km $\mathrm{s^{-1}}$; $\Delta\lambda_D = \lambda v_D/c = 44.7$ mÅ. (b) $\Delta\nu_D = v_D/\lambda = 3.86 \times 10^9$ Hz, so $a = \gamma/(4\pi\Delta\nu_D) = 0.031$. (c) Solve $e^{-v^2} = a/(\sqrt{\pi}v^2)$: $v \approx 2.41$, i.e. $2.41 \times 44.7 \approx 108$ mÅ from centre. Beyond about 0.1 Å the wings are controlled by collisions, which is why the broad Na D wings measure pressure.

Part C — Essay (choose one, 10 marks)

C1. "Local thermodynamic equilibrium is a statement about matter, not about radiation." Discuss, with reference to Kirchhoff's law, the two-level atom and the thermalisation depth.

Model answer outline. (1) Define TE and LTE; Maxwell, Boltzmann and Saha at the local T. (2) Kirchhoff: $j = \alpha B$ depends only on the matter state, hence $S = B$ in LTE while $I \neq B$ (escape at the surface; $I(0, \mu < 0) = 0$; $J(0) \approx B/2$). (3) Validity: collisions must dominate radiative rates; two-level atom $S = (1-\varepsilon)\bar{J} + \varepsilon B$, so LTE when $\varepsilon \to 1$. (4) Even with ε close to but below 1, S deviates from B within $\tau \sim 1/\sqrt{\varepsilon}$ of the surface, with $S(0) = \sqrt{\varepsilon} B$. (5) Consequences: LTE is good for continua and weak lines in dwarf photospheres; it fails for strong lines, chromospheres, hot or low-density stars and nebulae. (6) Practical: LTE turns the RTE into an explicit formal solution once $T(\tau)$ is known; NLTE requires solving rate equations coupled to the radiation field (ALI).

C2. Describe how a model stellar atmosphere is computed, starting from the gray LTE solution, and how its predictions are compared with observations.

Model answer outline. Inputs $T_{\rm eff}$, $\log g$, composition → gray starting $T(\tau) = [\frac{3}{4} T_{\rm eff}^4 (\tau + q)]^{1/4}$ → hydrostatic $P(\tau)$ from $dP/d\tau = g/\kappa$ → Saha–Boltzmann populations → κ_ν, j_ν → solve the RTE at many frequencies (formal solution, or Feautrier/ALI with scattering) → check radiative equilibrium $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$ and constant flux → temperature correction (Unsöld–Lucy) → iterate to convergence → emergent spectrum, limb darkening, line profiles (Milne–Eddington intuition, curve of growth) → compare with observed spectra, colours, Balmer jump and line equivalent widths → derive $T_{\rm eff}$, g and abundances.

Part D — Regimes and opacity (4 marks each)

D1. (Lectures SA1, SA3) Classify as TE, LTE or NLTE, with one sentence of justification each: (a) radiation 0.5 $R_\odot$ from the centre of the Sun; (b) the continuum at 500 nm in the solar photosphere; (c) the core of the Ca II K line; (d) [O III] 500.7 nm in a planetary nebula with $n_e = 10^{10}$ m^{-3}.

Solution. (a) **TE**: the photon mean free path (cm) is 10^{-10} of the temperature scale height, so the radiation is isotropic and Planckian at the local T. (b) **LTE**: H^- populations are set by collisions at $n_e \sim 10^{19}$–10^{20} m^{-3}, so $S = B$, although $I \neq B$ at the surface. (c) **NLTE**: a strong resonance line forming high in the low-density chromosphere, with photon losses ($S < B$) and non-radiative heating. (d) **NLTE**: $n_e \approx 0.016\, n_{\rm crit}$, so every collisional excitation is followed by radiative decay.

D2. (Lecture SA5) Explain line blanketing and backwarming, and name the dominant continuum opacity source in (a) an O star, (b) an A0 star, (c) the Sun, (d) an M dwarf.

Solution. Many lines block flux (mainly in the UV). The energy must still escape, so it is redistributed to other frequencies and partly returned to deeper layers, which become hotter (backwarming), while the outer layers cool. Opacity sources: (a) electron scattering and He bound-free; (b) H I bound-free (Balmer continuum); (c) H^- bound-free and free-free; (d) H^- plus molecules (TiO, H_2O).

Formula Sheet — Course B: Stellar Atmospheres

LTE (Lecture SA1)

- Kirchhoff: $j_\nu = \alpha_\nu B_\nu(T) \Rightarrow S_\nu = B_\nu(T)$ (but $I_\nu \neq B_\nu$)
- Boltzmann: $\dfrac{N_j}{N_i} = \dfrac{g_j}{g_i} e^{-\Delta E/kT}$ · Saha: $\dfrac{N_{r+1} n_e}{N_r} = \dfrac{2U_{r+1}}{U_r}\left(\dfrac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi/kT}$, $(2\pi m_e k/h^2)^{3/2} = 2.415\times10^{21}$ m^{-3}K$^{-3/2}$
- LTE opacity factor $1 - e^{-h\nu/kT}$ · two-level atom: $S = (1-\varepsilon)\bar{J} + \varepsilon B$, $\varepsilon' = \frac{C_{21}}{A_{21}}(1-e^{-h\nu/kT})$, $\varepsilon = \frac{\varepsilon'}{1+\varepsilon'}$ · $n_{\rm crit} = A_{21}/q_{21}$

Continuous opacity (Lecture SA2)

- Kramers: $\sigma_{\rm bf}(n,\nu) = 2.815\times10^{29} Z^4 g_{\rm bf}/(n^5\nu^3)$ cm^2 for $\nu \geq \nu_n = 3.288\times10^{15} Z^2/n^2$ Hz; edges $\lambda_n = 91.13\, n^2/Z^2$ nm
- LTE: $\alpha_\nu^{\rm bf} = \sum_n N_n \sigma_{\rm bf}(n,\nu)(1-e^{-h\nu/kT})$, $N_n = N_{\rm H\,I} 2n^2 e^{-E_n/kT}/U$
- Free–free: $\alpha_\nu^{\rm ff} = 3.69\times10^8 Z^2 n_e n_i T^{-1/2}\nu^{-3} g_{\rm ff}(1-e^{-h\nu/kT})$ cm^{-1} (cgs)
- H$^-$: $\chi = 0.754$ eV, threshold 1.64 µm, $\sigma_{\max} \approx 4\times10^{-17}$ cm^2 near 850 nm; $\frac{n(\mathrm{H^-})}{n(\mathrm{H\,I})n_e} = \frac{1}{4}\left(\frac{h^2}{2\pi m_e kT}\right)^{3/2} e^{0.754\,\mathrm{eV}/kT}$
- Thomson: $\sigma_T = 6.652\times10^{-29}$ m^2, $\kappa_{\rm es} = 0.020(1+X)$ m^2/kg · Rayleigh $\propto \lambda^{-4}$ · Kramers' law $\bar{\kappa} \propto \rho T^{-3.5}$

Transition rates (Lecture SA3)

- Statistical equilibrium: $n_i \sum_j (R_{ij} + C_{ij}) = \sum_j n_j (R_{ji} + C_{ji})$
- Photo-ionisation: $R_{i\kappa} = 4\pi \int \frac{\sigma_i}{h\nu} J_\nu d\nu$ · recombination (Milne): $R_{\kappa i} = \left(\frac{n_i}{n_\kappa}\right)^* 4\pi \int \frac{\sigma_i}{h\nu}\left(\frac{2h\nu^3}{c^2} + J_\nu\right) e^{-h\nu/kT} d\nu$
- Collisions: $q_{ul} = 8.629\times10^{-6}\Omega/(g_u\sqrt{T})$ cm^3 s^{-1} · Klein–Rosseland: $q_{lu} = (g_u/g_l) q_{ul} e^{-\Delta E/kT}$
- Critical density $n_{\rm crit} = A_{ul}/q_{ul}$ · two-level, no radiation: $\frac{(n_u/n_l)}{(n_u/n_l)^*} = \frac{n_e/n_{\rm crit}}{1+n_e/n_{\rm crit}}$
- Coronal: $n_u/n_l = n_e q_{lu}/A_{ul}$; $n_{i+1}/n_i = q_{\rm ion}(T)/\alpha_{\rm rec}(T)$ · Nebular: $J_\nu \approx W B_\nu(T_*)$, $W \approx \frac{1}{4}(R_*/r)^2$

Line absorption and broadening (Lecture SA2)

- $\alpha_\nu^{\rm line} = \frac{\pi e^2}{m_e c} f_{lu} n_l \phi(\nu)(1-e^{-h\nu/kT})$, $\frac{\pi e^2}{m_e c} = 0.02654$ cm^2 Hz · $A_{ul} = 0.6670 \frac{g_l}{g_u}\frac{f_{lu}}{\lambda^2[\mathrm{cm}]}$ s^{-1}
- Natural: $\gamma_{\rm rad} = \Gamma_u + \Gamma_l$, $\Delta\lambda_{\rm FWHM} = \lambda^2\gamma/(2\pi c)$; classical $\gamma_{\rm cl} = 2.47\times10^{-22}\nu^2$ s^{-1}
- Doppler: $\Delta\nu_D = \frac{\nu_0}{c}\sqrt{2kT/m + \xi^2}$ · pressure: $\Delta\nu = C_n/r^n$, $n = 2$ (linear Stark), 3 (resonance), 4 (quadratic Stark), 6 (van der Waals); impact width $\gamma_n \propto N C_n^{2/(n-1)} \bar{v}^{(n-3)/(n-1)}$
- Voigt: $\phi = H(a,v)/(\sqrt{\pi}\Delta\nu_D)$, $a = \gamma/(4\pi\Delta\nu_D)$, wings $H \approx a/(\sqrt{\pi}v^2)$ · rotation: $\Delta\lambda_L = \lambda v \sin i/c$

Gray LTE atmosphere (Lecture SA4)

- $J = B = S$, $H = \sigma T_{\rm eff}^4/4\pi$
- Eddington: $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$, $T(0) = 0.841 T_{\rm eff}$, $T(2/3) = T_{\rm eff}$
- Exact: $T^4 = \frac{3}{4}T_{\rm eff}^4[\tau + q(\tau)]$, $q(0) = 1/\sqrt{3}$, $q(\infty) = 0.7104$, $T(0) = 0.811 T_{\rm eff}$
- Limb darkening: Eddington $I(\mu)/I(1) = (2 + 3\mu)/5$; exact $H(\mu)/H(1)$, $I(0)/I(1) = 0.344$
- Two-stream ($\mu = \pm 1/\sqrt{3}$): $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + 1/\sqrt{3})$
- Hydrostatic: $dP/d\tau = g/\kappa \Rightarrow P = g\tau/\kappa$

Lines (Lecture SA4)

- $\int \sigma_\nu d\nu = \pi r_e c f \cdot \Delta\nu_D = \frac{\nu_0}{c}\sqrt{2kT/m + \xi^2}$ · Voigt $a = \gamma/4\pi\Delta\nu_D$
- Milne–Eddington (LTE): $\dfrac{F_\nu}{F_c} = \dfrac{1 + 2\beta/[3(1+\eta_\nu)]}{1 + 2\beta/3}$, maximum depth $1 - \frac{1}{1+2\beta/3}$
- Weak line: $W_\lambda = \pi r_e \lambda^2 N f$ · Curve of growth: $W \propto N \to \sqrt{\ln N} \to \sqrt{N}$
- NLTE: $S^{\rm line} = \frac{2h\nu^3}{c^2}\left[\frac{b_l}{b_u}e^{h\nu/kT} - 1\right]^{-1}$

TE, LTE, NLTE (Lectures SA1, SA3)

- TE: $I_\nu = B_\nu(T)$ · LTE: $S_\nu = B_\nu(T(\mathbf{r}))$ · NLTE: rate equations, $S_\nu \neq B_\nu$
- Statistical equilibrium: $n_i \sum_j (R_{ij} + C_{ij}) = \sum_j n_j(R_{ji} + C_{ji}) \cdot b_i = n_i/n_i^{\rm LTE}$
- $S^{\rm line} = \frac{2h\nu^3}{c^2}\left[\frac{b_l}{b_u}e^{h\nu/kT} - 1\right]^{-1} \approx \frac{b_u}{b_l}B_\nu$ · criterion: n_e vs $n_{\rm crit} = A\beta_{\rm esc}/q$

Stellar atmospheres (Lecture SA5)

- $dP/d\tau = g/\kappa \cdot F_{\rm rad} + F_{\rm conv} = \sigma T_{\rm eff}^4$ · scale height $H = kT/(\mu m_H g)$
- Convection if $\nabla_{\rm rad} > \nabla_{\rm ad}$ (0.4 ideal monatomic); gray atmosphere: $\nabla_{\rm rad} = \tau/[4(\tau + 2/3)] < 0.25$
- Hand-built model (Novotny Ch. 9): $T = T_{\rm eff}\,[\frac{3}{4}(\tau + \frac{2}{3})]^{1/4}$; for $\kappa = \kappa_0 P^a$: $P^{1+a} = (1+a)\,g\tau/\kappa_0$ (Sun: $\kappa \approx 10^{-4.4}P^{0.8}$ cgs); stable if $d\ln P/d\ln T > 1/\nabla_{\rm ad} = 2.5$